ScienceDirect

Journal of Manufacturing Processes

Date: 30 May 2026

Show more

Research article Full text access Get rights and content

Interpretable arc stability monitoring via Physics-Guided Data Programming in robotic gas metal arc welding

Lei Zhang [a b 1], Lin Chen [a 1], Binqi Jia [c], Sichuang Yang [a], Minling Pan [a], Haihong Pan [a] Show more

View PDF Download full issue Cite Add to Mendeley Share

10.1016/j.jmapro.2026.03.049

Article Recommended articles

Outline

Abstract

The advancement of intelligent monitoring for Gas Metal Arc Welding (GMAW) is critically constrained by the scarcity of high-fidelity labeled data and the prohibitive cost of manual annotation. To address this challenge, a Physics-Guided Data Programming (PGDP) framework is proposed to integrate domain-specific welding physics with weak supervision for scalable arc stability monitoring in robotic weaving GMAW. Instead of relying on manual labels, the framework leverages expert knowledge, calibrated via synchronized high-speed imaging, to encode multi-dimensional physical heuristics into programmatic Labeling Functions (LFs). A generative model then aggregates these potentially conflicting weak supervision signals to produce probabilistic training labels for a discriminative deep learning model. SHapley Additive exPlanations (SHAP)-based interpretability is used for feature optimization and model analysis. It reveals that high-order physical features, such as Power Kurtosis, serve as informative signatures of arc instability and are difficult to capture using conventional threshold-based criteria. Validated on a robotic weaving V-groove welding dataset, the proposed method consistently outperforms representative statistical and learning-based baselines, achieving an F1-score of 0.952 and demonstrating robust generalization across diverse operating conditions. Overall, this study establishes a cost-effective and data-efficient paradigm for industrial welding quality monitoring by systematically integrating physics-guided weak supervision with scalable data-driven learning.

Keywords

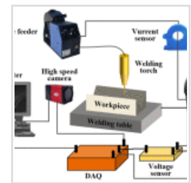
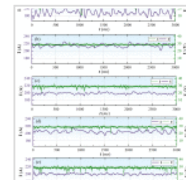
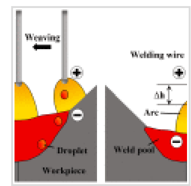
Gas metal arc welding; Data programming; Weak supervision; Welding arc stability

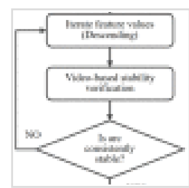
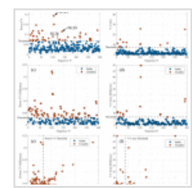
Previous article in this issue

1. Introduction

Gas Metal Arc Welding (GMAW) is one of the most widely adopted joining processes in modern manufacturing, owing to its high deposition efficiency, operational flexibility, and cost-effectiveness. With the rapid advancement of robotic systems and intelligent manufacturing paradigms, GMAW has been increasingly integrated into automated production lines across automotive, aerospace, and heavy equipment industries [1]. While robotic welding enables consistent torch positioning and repeatable motion control, achieving stable and defect-free welds remains a persistent challenge. The welding process involves highly coupled and nonlinear interactions among arc behavior, metal transfer, and molten pool dynamics. These interactions make the process inherently sensitive to disturbances and operating variations. As a result, perturbations in arc

Figures (15)

Show 9 more figures

Tables (7)

behavior and metal transfer can readily destabilize the process, leading to inconsistent heat input and degraded weld quality.

In industrial practice, weld quality assurance in robotic GMAW predominantly relies on post-weld inspection, specifically Non-Destructive Testing (NDT) techniques such as X-ray radiography [2], ultrasonic testing [3], and optical microscopy [4]. Although these methods provide high-resolution characterization of weld defects, they are inherently retrospective and incapable of capturing process instabilities as they evolve. As a result, unstable welding conditions may remain undetected throughout production and are only revealed during post-process inspection, leading to excessive scrap, rework, and delayed feedback. Moreover, the substantial capital investment and operational costs associated with NDT systems significantly hinder their scalability and widespread deployment in intelligent manufacturing scenarios. Collectively, these limitations underscore the necessity for a paradigm shift from reactive, offline inspection toward proactive, real-time monitoring of welding process stability.

To overcome the limitations of offline inspection, data-driven approaches for real-time weld monitoring have attracted considerable attention. Existing methods can be broadly categorized into supervised and unsupervised paradigms [5], each with inherent trade-offs. Supervised learning models can achieve high predictive accuracy but require large volumes of expert-labeled data, which are costly and often unavailable in industrial settings. By contrast, unsupervised methods avoid the labeling bottleneck but rely solely on statistical patterns in the data, without incorporating domain knowledge of welding physics. Consequently, these models are sensitive to typical process variability, limiting their robustness and practical reliability. This fundamental trade-off between scalability and physical interpretability motivates the development of hybrid frameworks that can leverage both data-driven learning and expert knowledge.

To bridge the gap between the scalability of data-driven learning and the interpretability of domain knowledge, a Physics-Guided Data Programming (PGDP) framework is proposed. Unlike pure data-driven or pure rule-based methods, the proposed framework synergizes domain-specific welding physics with weak supervision. Specifically, expert heuristics derived from high-speed imaging are encoded into programmatic Labeling Functions (LFs), enabling the generation of probabilistic labels at scale without manual annotation. Crucially, the framework incorporates post-hoc interpretability analysis to verify physical consistency, identifying dominant indicators such as Voltage and Power Coefficient of Variation differences. Consequently, the proposed framework enables scalable, annotation-efficient, and physically interpretable welding process monitoring, providing a practical solution for real-world industrial deployment.

The main contributions of this study are threefold:

(1) Physics-guided weak supervision for scalable label generation: The proposed framework enables large-scale probabilistic label generation by encoding expert physical heuristics as programmatic labeling functions and automatically denoising their conflicts, substantially reducing manual annotation effort and systematic bias.

(2) Interpretability-driven physical insight and optimization: Leveraging SHapley Additive exPlanations (SHAP)-based analysis, the framework not only guides physical feature reduction but also verifies the physical consistency of the learned patterns, uncovering high-order instability precursors (e.g., Power Kurtosis) that surpass the detection capabilities of conventional static heuristics.

(3) A cost-effective and generalizable monitoring framework: Integrating weak supervision with discriminative modeling, the proposed approach demonstrates strong robustness to label noise and superior generalization capability, enabling reliable detection of industrial arc instability patterns and providing practical decision support for real-world robotic welding applications.

The remainder of this paper is organized as follows. Section 2 reviews related work and identifies the existing research gap. Section 3 describes the experimental methodology, including the welding setup and data acquisition procedures. Section 4 presents the proposed PGDP framework in detail. Section 5 discusses the experimental results, and Section 6 concludes the paper.

2. Related work

2.1. Impact of arc stability on weld quality

Within the GMAW process, the stability of the electric arc plays a decisive role in determining weld quality. As the primary medium for energy transfer and metal deposition, arc behavior directly governs droplet transfer modes, heat input distribution, and molten pool dynamics. Stable arc operation promotes regular droplet detachment and uniform thermal cycles, which are essential for achieving dense microstructures and smooth bead morphologies. In contrast, arc instabilities—such as arc-length fluctuations, voltage oscillations, and irregular droplet transfer—disrupt the dynamic equilibrium of the welding process and induce cascaded effects across multiple physical domains. Reduced arc stability can impair gas shielding effectiveness, facilitating hydrogen entrapment and oxide formation that result in porosity defects [6]. Simultaneously, uncontrolled variations in heat input hinder molten metal spreading and fusion, leading to defects such as undercut, incomplete fusion, and insufficient penetration [7]. Furthermore, chaotic droplet dynamics generate excessive spatter and surface irregularities, degrading material utilization and introducing stress concentrators that compromise joint mechanical performance [8], [9]. Consequently, maintaining arc stability is critical for ensuring both metallurgical integrity and structural reliability in automated welding systems.

2.2. Sensors and signals for welding process monitoring

Effective monitoring of arc instability requires robust and reliable sensing strategies. Recent studies have increasingly adopted in-situ, multi-modal sensing approaches. Visual sensing techniques, such as high-speed imaging, and acoustic sensing methods have been widely explored to capture external manifestations of welding dynamics [10], [11]. However, in industrial environments, these modalities are often affected by arc glare, fumes, spatter, and ambient noise, which significantly degrade signal quality and reliability. In contrast, electrical signals, specifically welding current and voltage, are intrinsic descriptors of the welding process. These one-dimensional time-series signals directly reflect dynamic variations in arc length and droplet transfer behavior and are largely immune to external environmental disturbances. Owing to their robustness, low cost, and ease of acquisition, electrical signal-based monitoring remains one of the most practical and widely adopted solutions for real-time stability assessment in industrial robotic welding systems [12], [13]. Consistent with these industrial requirements, this research adopts electrical signals as the sole input modality to ensure robust monitoring performance under real-world conditions.

2.3. Existing approaches for process monitoring in arc-based processes

Existing approaches for process monitoring in arc-based processes can be broadly categorized into supervised, unsupervised, semi-supervised, and weakly supervised learning paradigms.

2.3.1. Supervised learning

Supervised learning formulates process monitoring as a fully labeled classification or regression problem, where discriminative models are trained to directly map sensor measurements to predefined stability or defect categories. Representative works span multiple sensing modalities, including electrical-signal-based regression using multilayer perceptrons for bead geometry prediction [12], vision-based defect localization through CNN-YOLO architectures [10], and acoustic-electrical feature modeling with wavelet scattering networks for porosity identification [11]. Although these approaches often achieve high predictive accuracy under controlled conditions, they share a common dependency on large volumes of precisely annotated data. In practical manufacturing environments, such annotations are costly and require substantial domain expertise. Moreover, supervised models typically operate under closed-set assumptions and suffer from severe class imbalance due to the rarity of anomalous events. These limitations significantly restrict their scalability, adaptability, and robustness in real-world deployment.

2.3.2. Unsupervised learning

To alleviate the heavy reliance on labeled data, unsupervised learning models characterize the intrinsic distribution of process signals and identify deviations from normal operating states. Typical strategies include clustering, manifold learning, and reconstruction-based anomaly detection. In arc visual monitoring, video-based anomaly networks and two-stage detection frameworks have been developed to capture spatiotemporal irregularities without prior labels [14], [15]. For one-dimensional process signals—including current, voltage, and acoustic emissions—clustering methods such as K-means [16] and reconstruction models such as frequency-informed convolutional autoencoders [17], [18] learn the manifold of stable processes and flag anomalies through reconstruction errors. While these methods offer improved scalability and label efficiency, they primarily rely on statistical regularities rather than explicit domain knowledge. Consequently, the detected anomalies often lack clear physical interpretation, making it difficult for operators to associate model outputs with specific welding mechanisms or actionable process adjustments. This “black-box” nature limits their practical reliability in industrial settings.

2.3.3. Semi-supervised learning

Semi-supervised learning aims to balance data efficiency and discriminative capability by combining a small set of labeled samples with abundant unlabeled data. Instead of merely detecting deviations, these methods attempt to construct decision boundaries for specific defect or stability categories. In the field of welding defect monitoring, representative strategies include GAN-based data augmentation to mitigate class imbalance [19], spectral or time–frequency representations for soft boundary learning in pulsed processes [20], and transfer learning schemes that leverage large unlabeled datasets for representation pre-training [21]. Although such approaches reduce labeling requirements and improve generalization compared with fully supervised models, their learning mechanisms remain predominantly data-driven. Domain knowledge is typically incorporated only implicitly through handcrafted features rather than explicitly encoded constraints. As a result, the interpretability and physical consistency of the learned decision rules remain limited.

2.3.4. Weak supervision

To explicitly integrate expert knowledge while maintaining scalability, weak supervision —particularly the Data Programming (DP) paradigm—has recently emerged as a promising alternative. Instead of manual annotation, DP enables domain experts to encode heuristic rules or physical insights into programmatic Labeling Functions (LFs), which generate noisy but informative supervision signals. A generative model then reconciles these potentially conflicting rules to produce probabilistic labels for downstream discriminative training [22], [23]. This paradigm has demonstrated effectiveness across several domains and has begun to show potential in manufacturing applications, such as anomaly detection in resistance welding processes [24]. By transforming expert knowledge into scalable supervision, weak supervision uniquely bridges the gap between data-driven learning and domain-informed reasoning. Nevertheless, its application to arc-based welding monitoring remains largely unexplored, particularly for continuous stability assessment using physics-consistent features.

A summary of the representative state-of-the-art methods discussed above is provided in Table 1.

Table 1. Representative learning paradigms for process monitoring in welding and related manufacturing processes.

Learning paradigm	**Ref.**	**Core method**	**Signal modality**	**Supervision requirement**	**Physics integration level**	**Primary challenge addressed**
Supervised	[12]	MLP Regressor	Electrical	Extensive labeled data	Limited (data-driven)	Accurate regression under fully labeled conditions
	[10]	YOLO-Attention	Visual	Dense pixel-level annotation	Limited (data-driven)	High-precision surface defect detection
	[11]	Wavelet Network	Acoustic + electrical	Balanced labeled samples	Moderate (feature-level)	Feature-informed porosity prediction
Unsupervised	[14]	Video Anomaly Network	Visual	None	Limited (statistical)	Detection of spatiotemporal anomalies
	[16]	K-Means Clustering	Acoustic	None	Limited (statistical)	Clustering of deposition states
	[18]	FICAE (Autoencoder)	Electrical	None (reconstruction-based)	Limited (statistical)	Deviation detection from stable process manifold

Learning paradigm	Ref.	Core method	Signal modality	Supervision requirement	Physics integration level	Primary challenge addressed
Semi-supervised	[19]	StyleGAN + Boosting	Visual	Sparse + unlabeled	Limited (generative)	Mitigation of severe class imbalance
	[20]	Spectral Semi-supervised Learning	Electrical	Sparse + unlabeled	Moderate (spectral features)	Soft boundary learning for stability states
	[21]	Deep Transfer Learning	Visual	Sparse + unlabeled	Limited (data-driven)	Knowledge transfer under limited supervision
Weak supervision	[24]	Data Programming	Resistance signals	Heuristic rules	High (rule-based)	Scalable labeling for discrete welding events
	The proposed	PGDP	Electrical	Physics-guided heuristics	High (explicit physics constraints)	Scalable monitoring of continuous arc dynamics

2.4. Limitations of existing studies and research gap

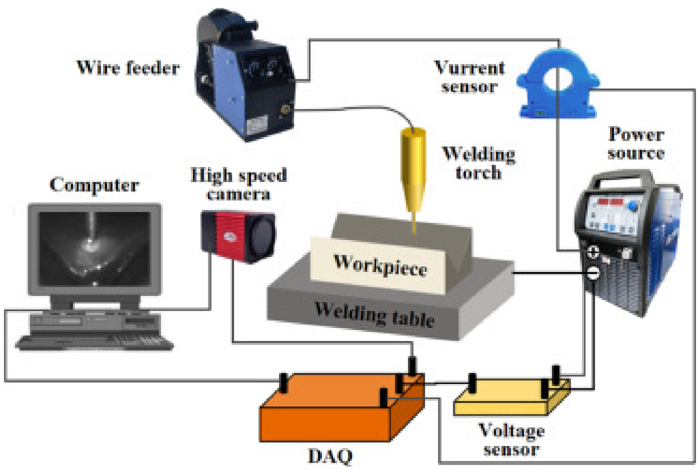
While the aforementioned approaches have advanced the state of the art in specific scenarios, a fundamental trade-off remains unresolved: existing methods struggle to achieve scalability in label generation while simultaneously maintaining physical interpretability in diagnostic reasoning. Unsupervised and semi-supervised models are effective at identifying statistical deviations but typically fail to explicitly reveal the physical origins of arc instability, resulting in diagnostic ambiguity. Conversely, applying weak supervision to arc welding is challenged by the difficulty of constructing robust, physics-consistent labeling rules. Arc stability is governed by tightly coupled multi-physical dynamics involving electrical, thermal, and geometric interactions. Simple threshold-based heuristics are often too rigid to capture subtle non-linear precursors, whereas overly complex rules risk overfitting and poor generalization. Consequently, a systematic framework that can extract salient physical features and translate expert knowledge into robust, physics-guided labeling functions—while maintaining scalability and interpretability—remains an open challenge. This work directly addresses this gap by proposing a Physics-Guided Data Programming framework tailored for industrial GMAW arc stability monitoring.

3. Experimental methodology

3.1. Experimental platform

Experiments were conducted on an automated Gas Metal Arc Welding (GMAW) platform comprising a digital power source, a wire feeder, a robotic arm for torch manipulation, and a shielding gas delivery unit. Q235 mild steel plates with a 90° V-groove served as the base material, while ER50–6 solid wire with a diameter of 1.2 mm was used as the filler metal. A shielding gas mixture of 80% Ar and 20% CO_2 was supplied at a constant flow rate.

The welding process was monitored using an integrated data acquisition system. Electrical signals were captured using a Hall-effect current sensor and a voltage sensor, digitized at a sampling rate of 1 kHz by a data acquisition (DAQ) card. The selection of this sampling frequency is grounded in the specific dynamic characteristics of the standard constant voltage GMAW process employed in this study. Operating primarily in the spray transfer regime, the electrical signals exhibit a quasi-stationary nature, where the dominant frequencies associated with natural droplet transfer and stability perturbations typically fall below 200 Hz. Consequently, the 1 kHz sampling rate (corresponding to a Nyquist frequency of 500 Hz) provides sufficient bandwidth to capture the relevant feature dynamics without aliasing. Synchronized with the DAQ, a high-speed camera recorded arc behavior and droplet transfer dynamics at 2000 frames per second (fps), ensuring sufficient temporal resolution for correlating visual observations with electrical signals. A schematic of the complete experimental setup is shown in Fig. 1.



Download: Download high-res image (113KB)

Download: Download full-size image

Fig. 1. The experimental platform.

To construct a comprehensive dataset encompassing diverse stability regimes, representative of actual industrial production, a V-groove root pass weave welding process was employed. This specific configuration is widely utilized in critical sectors such as pipeline construction and heavy machinery fabrication. This technique uses periodic torch oscillation to bridge the groove gap, ensuring uniform sidewall fusion and complete penetration at the joint base through controlled heat distribution. To facilitate accurate synchronization between arc images and electrical signals, unobstructed visibility for the high-speed camera was maintained throughout the experiments. A systematic experimental design was implemented to replicate distinct stability behaviors by modulating three critical process parameters: torch dwell time, welding speed, and weave amplitude. This multi-parameter approach captures authentic process deviations encountered in industrial practice, thereby enabling a rigorous investigation of arc stability under varying energetic and geometric constraints while ensuring reliable correlation between image and signal data.

As detailed in Table 2, five distinct experimental conditions were established to evaluate the framework's generalization capability. Condition 2 serves as the nominal baseline with optimized parameters, representing a stable process window. Conditions 1 and 3 modulate localized heat input at the groove sidewalls by varying dwell time (0.2 s and 0.6 s), inducing instability through insufficient or excessive pool elevation. Additionally, Condition 4 reduces welding speed to 5 mm/s to alter linear heat input, testing the response to high-energy deposition, while Condition 5 decreases weave amplitude to 2 mm to modify the arc scanning area, assessing stability under tighter geometric constraints. The welding arc signals under the five conditions are presented in Fig. 2.

Table 2. The welding process parameters for the five conditions.

Condition	**Wire feed speed (m/min)**	**Welding speed (mm/s)**	**Weave amplitude (mm)**	**Dwell time (s)**
1	10	7	4	0.2
2	10	7	4	0.4
3	10	7	4	0.6
4	10	5	4	0.4
5	10	7	2	0.4

Note: The welding current and arc voltage are reference values under the synergistic mode; actual values fluctuate dynamically with the arc state.

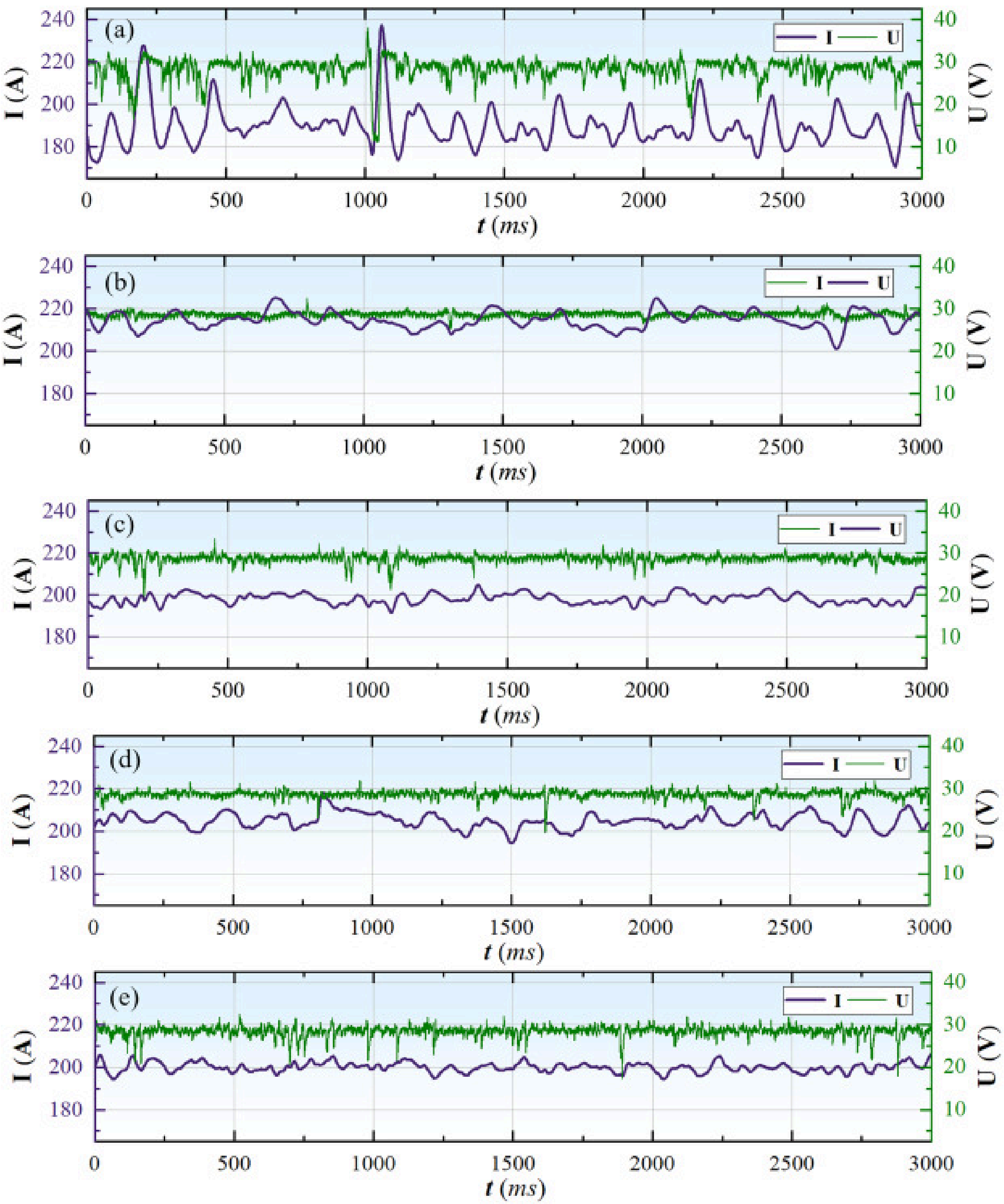

(a)
(b)
(c)
(d)
(e)
I
U
I (A)
U (V)
t (ms)
240
220
200
180
40
30
20
10
0
500
1000
1500
2000
2500
3000

Fig. 2. Welding arc signals under the five experimental conditions (Table 2): (a) Condition 1 (dwell time: 0.2 s); (b) Condition 2 (dwell time: 0.4 s); (c) Condition 3 (dwell time: 0.6 s); (d) Condition 4 (welding speed: 5 mm/s); (e) Condition 5 (weave amplitude: 2 mm).

3.2. Formation mechanism of defect

A critical determinant of arc stability in V-groove weave welding is the height of the molten pool, which constitutes the dynamic component of the effective cathode in conjunction with the static groove geometry. This molten pool height is primarily dictated by the torch dwell time, welding speed, and weave amplitude through its regulation of localized heat input, thereby influencing the adaptive behavior of the arc. The mechanism of arc instability is schematically illustrated in Fig. 3.

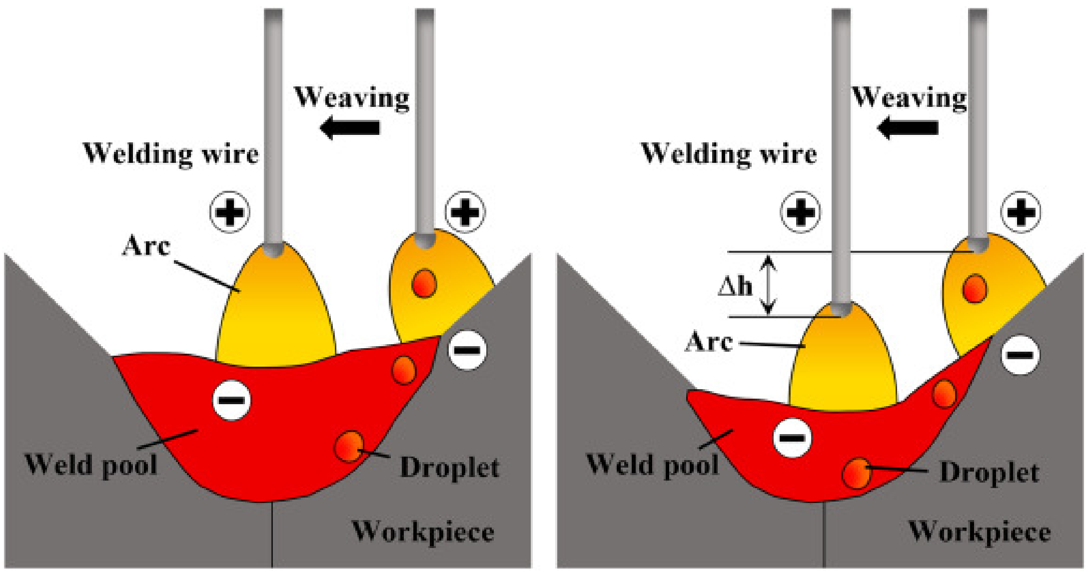


Fig. 3. Schematic diagram of the mechanism of unstable arc in V-shaped groove welding.

Deviations from optimal parameter windows disrupt this equilibrium through distinct physical mechanisms. Insufficient heat deposition, typified by insufficient dwell time, results in an underdeveloped molten pool, forcing rapid arc elongation that exceeds the system's self-regulation capacity. Conversely, excessive heat accumulation—whether induced by prolonged dwell time, reduced welding speed (increased linear heat input), or constricted weave amplitude (concentrated thermal load)—causes the molten pool to over-elevate or oscillate violently. This geometric distortion interferes with droplet transfer dynamics, promoting instability manifested as erratic short-circuiting, large-scale pool turbulence, or explosive spatter.

Crucially, these dynamic perturbations are not merely transient fluctuations but serve as direct precursors to irreversible weld defects. The induced molten-pool turbulence compromises the shielding gas envelope, exacerbating hydrogen absorption and accelerating oxide entrapment, which collectively promote porosity formation. Simultaneously, the resultant asymmetric thermal distribution along the groove walls impairs local wettability and hinders proper sidewall fusion, leading to defects such as incomplete fusion, undercut, and toe-cracking [25]. These causal links between signal fluctuations and defect formation underscore the necessity of arc-stability monitoring, as it enables the identification of potential failure precursors before geometric imperfections become visually manifest.

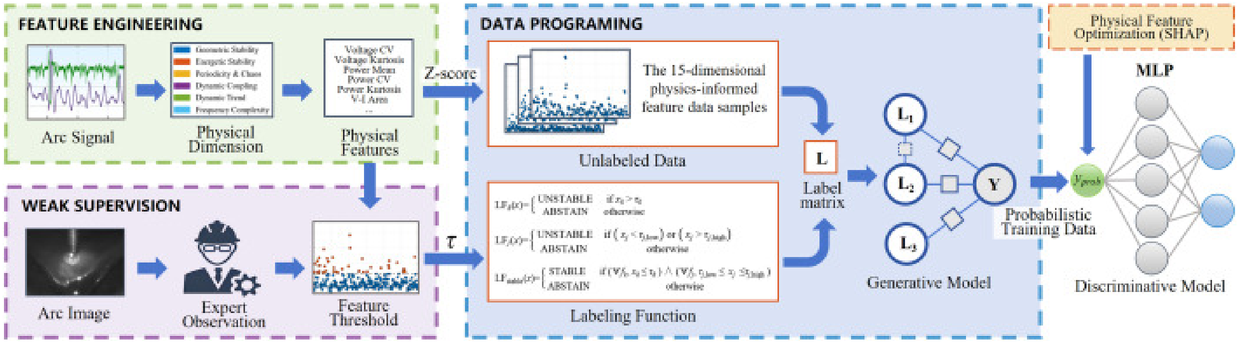
Therefore, understanding these physical pathways forms the foundation of the monitoring strategy developed in this work: the electrical signal signatures captured during arc instability—such as voltage surges, current dips, and short-circuit pulsations—directly correspond to the defect-formation mechanisms outlined above. This establishes a clear physics-based linkage between defect mechanisms and electrical-signal–based monitoring.

3.3. Signal preprocessing and segmentation

To establish a reliable dataset for stability analysis, a 3000 ms segment was extracted from the middle phase of the welding process for each experimental condition. The raw voltage and current signals were first denoised using a 4th-order Butterworth low-pass filter with a cutoff frequency of 480 Hz to suppress high-frequency noise and prevent aliasing. The filtered continuous signal was then segmented into consecutive, non-overlapping 10 ms windows, yielding 300 discrete temporal instances for analysis per condition. This window duration was specifically selected to encapsulate the characteristic periodicity of the spray transfer regime, where the natural droplet detachment frequency typically ranges from 100 to 200 Hz. By capturing these fundamental transfer events, the statistical features extracted from each window quantitatively represent the instantaneous degree of welding arc stability, providing a robust basis for the subsequent feature engineering and model training.

4. The Physics-Guided Data Programming (PGDP) framework

This section presents the architecture of the Physics-Guided Data Programming (PGDP) framework, as illustrated in Fig. 4. To effectively integrate domain knowledge with data-driven learning, the framework is structured into three sequential stages. First, Physics-Informed Feature Engineering extracts multi-dimensional physical feature vectors from raw electrical signals to quantitatively characterize arc dynamics. Second, Weak Supervision Generation applies Data Programming to encode expert heuristics into probabilistic training labels, enabling scalable supervision without manual annotation. Third, Discriminative Modeling and Optimization trains a deep learning model on these generated labels, followed by SHAP-based interpretability analysis to identify the most influential features and guide a rational reduction of the feature space, improving model efficiency while maintaining physical consistency.



Download: Download high-res image (586KB)

Download: Download full-size image

Fig. 4. The Physics-Guided Data Programming (PGDP) Framework.

4.1. Stage I: physical feature engineering

The first stage of the PGDP framework transforms raw electrical signals into a set of physical features for quantifying arc stability. To ensure a robust initial analysis, feature threshold calibration and preliminary model development leveraged data from Condition 1. This condition was selected as its parameters reliably generate a wide spectrum of arc dynamics, including significant unstable states. The resulting balanced dataset is crucial for the learning algorithm to acquire a generalized representation of instability.

From each 10ms time window, a multi-dimensional feature vector was systematically extracted. Each feature captures a distinct physical dimension of arc stability:

Geometric Stability: This dimension quantifies the stability of the arc's physical geometry and length. It incorporates the Voltage Coefficient of Variation, defined as the ratio of the standard deviation to the mean of the voltage signal, where higher values indicate greater instability. In addition, Voltage Kurtosis, a fourth-order statistical moment, is employed to quantify the impulsiveness of the voltage signal, thereby rendering it highly sensitive to sharp, transient spikes associated with phenomena such as arc re-ignition.

Energetic Stability: This dimension evaluates the consistency of energy delivery to the workpiece. It is characterized by the Power Mean, which reflects the average energy transfer rate, and the Power CV, which measures its relative fluctuation. To capture impulsive energy events, Power Kurtosis (representing the “tailedness” of the energy distribution) and Power Crest Factor (representing the “peakiness” of the distribution) are also included. Both indicators are particularly responsive to the abrupt energy bursts that occur during short-circuiting and spatter events.

Periodicity and Chaos: From the perspective of nonlinear dynamics, a stable arc is fundamentally periodic. The V—I Area, defined as the net energy consumed per cycle and enclosed by the V—I dynamic characteristic loop, is used to assess cycle consistency. Furthermore, the difference in V—I Area between consecutive loops directly quantifies non-periodic, chaotic behavior, with large deviations indicating the onset of chaos [26].

Dynamic Coupling: This dimension employs electrical resistance as a proxy for the physical state of the welding arc. The Resistance Mean serves as a fingerprint distinguishing high-resistance arcing from low-resistance short-circuiting states. In addition, the Resistance Range, defined as the difference between the maximum and minimum resistance, directly quantifies the magnitude of state transitions, which becomes pronounced under instability involving frequent short-circuits.

Dynamic Trend: This dimension captures the temporal evolution of arc states to provide predictive capability. It includes Voltage CV Difference and Power CV Difference, which represent the rates of change in voltage and power variability across consecutive windows, thereby signaling rapid state transitions. In addition, Power Mean Difference reflects the change in mean power between windows, effectively describing the acceleration of energy delivery.

Frequency Complexity: This dimension evaluates randomness in the frequency domain through Voltage Spectral Entropy and Current Spectral Entropy. Both features apply Shannon entropy to the Power Spectral Density (PSD), where higher entropy reflects a uniform, noise-like spectrum characteristic of chaotic instability, whereas lower entropy signifies a periodic and predictable process [27].

This comprehensive set of 15 features was established based on preliminary experiments and a systematic review of the existing literature, as summarized in Table 3. Although broad in scope, the feature set is likely to contain redundancy; therefore, subsequent stages of PGDP incorporate physical feature optimization to identify the most salient and non-redundant features of arc instability.

Table 3. Summary of the selected physical features.

Physical dimension	Feature name	Symbol
Geometric stability	Voltage CV	$C_{v,U}$
	Voltage kurtosis	k_U
Energetic stability	Power mean	μ_P
	Power CV	$C_{v,P}$
	Power kurtosis	k_{P}
	Power crest factor	C_P
Periodicity & chaos	V-I area	A_{VI}
	V-I area difference	ΔA_{VI}

Physical dimension	**Feature name**	**Symbol**
Dynamic coupling	Resistance mean	μ_R
	Resistance range	ΔR
Dynamic trend	Voltage CV difference	$\nabla C_{v,U}$
	Power CV difference	$\nabla C_{v,P}$
	Power mean difference	$\Delta \mu_P$
Frequency complexity	Voltage spectral entropy	$H_{SE,U}$
	Current spectral entropy	$H_{SE,I}$

4.2. Stage II: weak supervision generation via data programming

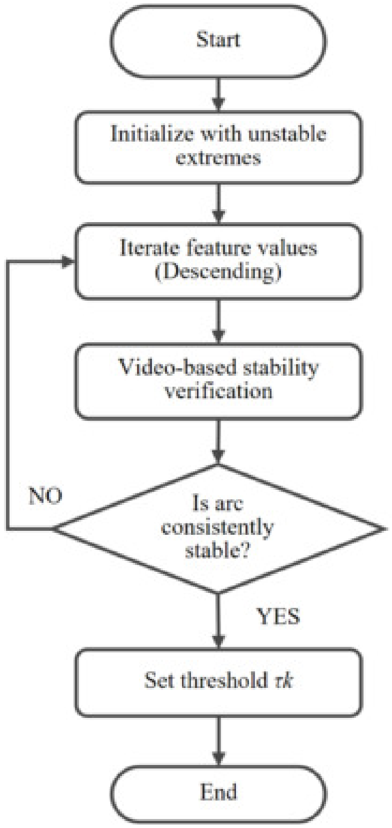
High-fidelity arc stability labels are scarce owing to the high cost of synchronized videography and manual annotation. To implement the Physics-Guided strategy, Data Programming [23] was employed to encode expert-derived physical heuristics into probabilistic supervision signals. The procedure comprises three steps.

4.2.1. Calibration of physical thresholds

A foundational step in Data Programming was establishing quantitative thresholds for the engineered features. These physically-grounded thresholds formed the logical basis for all heuristic labeling functions (LFs).

Calibration was performed by welding experts through a systematic analysis of feature scatter plots alongside synchronized high-speed video. A binary classification scheme (0: STABLE/1: UNSTABLE) was adopted. An intermediate “transitional” class was specifically avoided. Such definitions are inherently subjective and provide limited value for industrial monitoring, where unambiguous detection of instability is required.

For each feature, calibration was performed through a structured expert-driven procedure. Experts first identified regions in the feature scatter plots associated with extreme values, which corresponded to severe arc instability as verified by synchronized high-speed video. From these unstable regions, feature values were examined progressively in descending order. At each step, the corresponding video segment was reviewed to assess whether the arc behavior had transitioned to a consistently stable state. The feature value at which the entire observed interval exhibited stable arc behavior was defined as the physically grounded stability threshold (τk). An overview of the calibration procedure is provided in Fig. 5.



Download: Download high-res image (146KB)

Download: Download full-size image

Fig. 5. The threshold calibration flowchart.

Finally, to eliminate scale-related bias, the entire feature space was normalized to the [0,1] range. Each expert-defined threshold (τk) underwent the same transformation. This ensured that the derived rules remained physically interpretable within a standardized feature space.

4.2.2. Construction of physics-based labeling functions

A labeling function (LF) is formally defined as a programmatic encapsulation of a single expert heuristic. In this study, three types of physics-based LFs were developed to reflect the diverse relationships between physical features and arc instability.

One-Sided Threshold LFs: This category applies to features that exhibit a monotonic relationship with instability. For example, Power Kurtosis and V—I Area Difference are typically associated with increasingly severe transient disturbances or process aperiodicity as their values rise. For a given feature x_k and its expert-defined threshold τ_k, the LF is defined as:

$$\mathrm{LF}_k(x)=\begin{cases} \mathrm{UNSTABLE} & \text{if } x_k > \tau_k \\ \mathrm{ABSTAIN} & \text{otherwise} \end{cases} \tag{1}$$

Two-Sided Range LFs: This category is suitable for features that possess a well-defined nominal operating range, such as Power Mean. Insufficient power may indicate an impending arc extinction, whereas excessive power may result in excessive penetration or unstable metal transfer modes. Consequently, deviations from the nominal range are regarded as anomalous. For a given feature x_j, with an expert-defined acceptable interval$[\tau_{j,low},\tau_{j,high}]$, the LF is defined as:

$$\mathrm{LF}_j(x)=\begin{cases}\text{UNSTABLE} & \text{if } (x_j<\tau_{j,\mathrm{low}}) \text{ or } (x_j>\tau_{j,\mathrm{high}})\\ \text{ABSTAIN} & \text{otherwise}\end{cases} \tag{2}$$

Stability-Confirming LF: To establish a strong positive indicator of stability, a single holistic LF was constructed. This function adopts a deliberately conservative logic, assigning the label STABLE only when all known indicators of instability are absent. The LF integrates the principles of the two categories described above:

$$\mathrm{LF}_{\mathrm{stable}}(x)=\begin{cases}\text{STABLE} & \text{if}(\forall f_k, x_k\le\tau_k)\wedge(\forall f_j, \tau_{j,\mathrm{low}}\le x_j\le\tau_{j,\mathrm{high}})\\ \text{ABSTAIN} & \text{otherwise}\end{cases} \tag{3}$$

Collectively, these LFs, comprising complementary rule types, generate a diverse set of weak supervision signals. When applied systematically to the training dataset, the complete suite produces a label matrix $\boldsymbol{L}$, which consolidates the raw, multi-dimensional expert heuristics. This matrix serves as the sole input to the subsequent generative model, where the individual votes are aggregated and de-noised to yield refined supervision signals.

4.2.3. Probabilistic label aggregation via a generative model

The labeling functions (LFs) produced in the previous stage generated a large volume of weak supervision signals that are inherently sparse and potentially conflicting. The outputs of all 16 LFs on the unlabeled training dataset were organized into a label matrix $\boldsymbol{L}\in\{-1,0,1\}^{n\times16}$, where n denotes the number of samples and the majority of entries correspond to ABSTAIN (−1). To systematically reconcile these noisy and incomplete votes, a generative model was employed to represent the structure of this label matrix.

In this study, the LabelModel from the Snorkel framework [23] was adopted, as it is widely utilized in weak supervision research. The LabelModel is a probabilistic graphical model designed to jointly estimate two unknown quantities without requiring access to ground-truth labels: (1) the accuracy of each LF, and (2) the dependency structure among

LFs. Unlike methods that assume independence, the LabelModel explicitly accounts for correlations by learning a covariance structure. Its parameters θ are estimated by maximizing the marginal log-likelihood of the observed label matrix $\boldsymbol{L}$, expressed as:

$$\theta^* = arg\max_{\theta} \log P_\theta(\boldsymbol{L}) = arg\max_{\theta} \log \sum_{\mathbf{Y}} P_\theta(\boldsymbol{L}|\boldsymbol{Y})\, P_\theta(\boldsymbol{Y}) \tag{4}$$

where θ^* denotes the optimal parameters, $\boldsymbol{L}$ is the observed label matrix, and $\boldsymbol{Y}$ is the latent true label vector. The marginalization over all possible $\boldsymbol{Y}$ ensures that the parameters are fitted to maximize the likelihood of the observed LF outputs.

Once trained, the generative model aggregates the LF votes to infer the most probable label distribution for each data point. It produces a probabilistic label $y_{prob,i} \in [0,1]$ for each sample x_i, representing the estimated probability that the sample belongs to the UNSTABLE class. This procedure reconciles conflicting LF outputs and synthesizes them into a unified supervisory signal. The resulting probabilistic labels are subsequently employed to supervise the discriminative model.

Three domain experts independently annotated these samples based strictly on high-speed video observations, while the model relied solely on electrical signal features. This cross-modal independence guarantees that the validation corroborates the physical correlation between signal fluctuations and process instability. Furthermore, to mitigate subjective variability, a blind majority voting strategy was adopted to reconcile individual expert annotations into a unified, objective benchmark for evaluation. This ground-truth dataset served exclusively for evaluating the quality of the probabilistic labels and was not involved in model training.

4.3. Stage III: discriminative modeling and interpretability-driven optimization

With the availability of a large programmatically labeled dataset, the subsequent stage of the PGDP framework focused on training a discriminative model capable of learning the complex, non-linear mappings from the high-dimensional physical feature space to the stability labels.

4.3.1. Discriminative model training

For this stage, the input consisted of normalized 15-dimensional feature vectors forming the input matrix X_{train}, with their corresponding probabilistic labels (y_{prob}) serving as the supervisory signal. The dataset was partitioned into training (80%) and validation (20%) subsets to enable monitoring of potential overfitting. Model selection was conducted via 5-fold cross-validation on the training data. The MLP was selected due to its favorable generalization ability, its inductive bias consistent with the underlying physical processes, and its compatibility with gradient-based feature attribution methods required for subsequent interpretability analyses.

The selected MLP architecture comprised two hidden layers with 128 and 64 neurons, respectively. Each hidden layer was followed by a Batch Normalization layer to stabilize training and a Dropout layer with a rate of 0.5 for regularization. The output layer contained a single neuron designed for binary classification, outputting a logit score mapped to the stability probability. The resulting model, hereafter referred to as PGDP-15, was trained on the complete 15-dimensional feature set. Training utilized the probabilistic labels (y_{prob}) generated by the preceding generative model, with Binary Cross-Entropy with Logits Loss adopted as the objective function. This loss function is particularly well-suited for probabilistic supervision, as it directly incorporates the confidence information derived from weak labeling. Optimization was performed using Adam with a learning rate of 0.001, for 100 epochs, with the data processed in shuffled mini-batches of size 32.

4.3.2. Feature optimization via SHAP analysis

A central objective of the proposed framework is to achieve both predictive accuracy and model interpretability. To this end, SHAP (SHapley Additive exPlanations) [28], a state-of-the-art method grounded in cooperative game theory, was employed. SHAP attributes a model's prediction to its input features by computing Shapley values, which represent the average marginal contribution of each feature across all possible feature coalitions. The explanation is formalized as an additive feature attribution model:

$$g\left(z'\right) = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i \tag{5}$$

where g denotes the explanation model, $z' \in \{0,1\}^M$ is a binary vector indicating the inclusion or exclusion of features in a coalition, M is the total number of input features, and ϕ_i is the Shapley value corresponding to feature i.

SHAP was applied to the trained PGDP-15 model to obtain a quantitative ranking of global feature importance. The ranking was derived by averaging the absolute Shapley values of each feature across the dataset. This analysis provides a data-driven assessment of the relative contribution of features, complementing and extending qualitative physical interpretations. Based on the SHAP ranking, the initial feature set was pruned to retain only the most influential predictors, and a final multilayer perceptron (MLP) was trained on this optimized subset. This iterative refinement not only enhances interpretability by concentrating on salient physical indicators but also yields a more parsimonious and computationally efficient model with reduced susceptibility to overfitting.

4.4. Evaluation metrics

To rigorously assess the performance of the proposed framework, a comprehensive set of metrics tailored for binary classification was employed. The selected metrics include Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC). Collectively, these metrics enable a holistic evaluation of predictive accuracy, robustness, and class-wise discrimination capability.

The F1-Score, defined as the harmonic mean of Precision (P_c) and Recall (R_c), provides a balanced evaluation of model performance under class-imbalanced conditions:

$$F1_c = 2 \bullet \frac{P_c \bullet R_c}{P_c + R_c} \tag{6}$$

The AUC metric offers a threshold-independent measure of the model's ability to distinguish between stable and unstable conditions, thereby reflecting both predictive quality and the reliability of programmatically generated labels. This comprehensive evaluation protocol facilitates an in-depth validation of the model's generalizability and operational reliability.

4.5. Baseline methods

To rigorously evaluate the proposed PGDP framework, seven representative baseline methods were implemented, covering statistical, unsupervised, and supervised paradigms commonly used for arc instability detection. All methods were evaluated on the same electrical-signal feature space and identical hold-out test set to ensure fair comparison. A detailed summary is provided in Table 4.

Table 4. Comparison of baseline methods and the proposed PGDP framework.

Method	Algorithm class	Supervision strategy	Physics integration	Key mechanism
3-Sigma Control Chart [24]	Statistical	Unsupervised	Indirect (feature selection)	Univariate thresholding
PCA [11], [29]	Statistical	Unsupervised	None	Linear subspace projection
Isolation Forest [24], [29]	Ensemble tree	Unsupervised	None	Random partitioning
LOF [24]	Density-based	Semi-supervised (one-class)	None	Local density estimation
One-Class SVM [30]	Kernel-based model	Semi-supervised (one-class)	None	Hyper-sphere boundary
Logistic Regression [11]	Linear model	Supervised	None	Linear classification

Method	Algorithm class	Supervision strategy	Physics integration	Key mechanism
Random Forest [24], [29]	Ensemble tree	Supervised	None	Bagging & decision trees
PGDP	Deep learning	Weakly supervised	Explicit (physics-guided LFs)	End-to-end non-linear mapping

The first group includes statistical and unsupervised approaches that operate without explicit labels and detect anomalies based on distributional or density deviations. Classical statistical baselines include a univariate 3-Sigma control chart [24] applied to the voltage coefficient of variation and multivariate PCA [11], [29] using reconstruction error. For non-parametric anomaly detection, Isolation Forest [24], [29] was trained on the full dataset, while Local Outlier Factor (LOF) [24] and One-Class SVM (OC-SVM) [30] were configured as novelty detectors using only stable arc samples to model the boundary of stable operations.

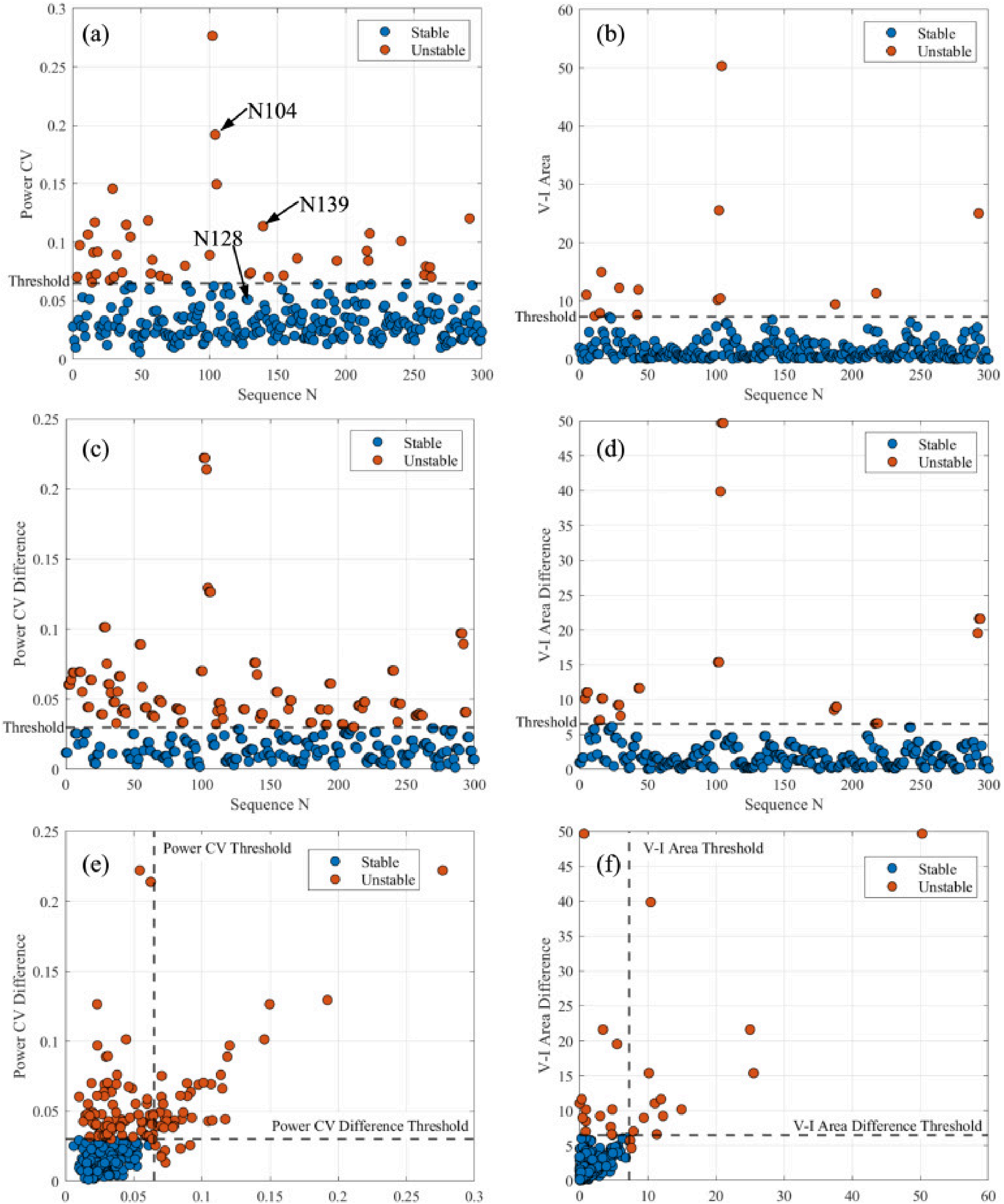
The second group comprises supervised classifiers to isolate the effect of model architecture from weak supervision. Logistic Regression [11] and Random Forest [24], [29] were adopted as representative linear and ensemble-based models. All supervised baselines were trained using the same probabilistic labels generated by the Label Model and evaluated against ground-truth annotations on the hold-out test set.

The proposed PGDP framework was implemented on a workstation equipped with two Intel Silver 4108 CPUs and an NVIDIA RTX 4080 Super GPU to enable GPU-accelerated training of the models. The Data Programming pipeline was constructed using the Snorkel library. Deep learning models were developed with the PyTorch framework, while cross-validation was conducted using Scikit-Learn. Model interpretability was achieved through SHAP analysis, implemented via the official SHAP library.

5. Results and discussion

5.1. Analysis of labeling function heuristics

A prerequisite for the Data Programming paradigm is the establishment of effective, physically interpretable heuristics. The labeled dataset analysis reveals a strong correlation between the proposed feature set and the physical state of the welding arc. As shown in Fig. 6, feature sequences can be clearly segmented into stable and unstable regions according to expert-defined thresholds (Section 4.2.1). The upper region corresponds to unstable values, while the lower region concentrates stable arc states. This distinct separation demonstrates that the selected features capture essential physical mechanisms of arc behavior.


(a)
Stable
Unstable
N104
N139
N128
Threshold
Power CV
Sequence N
(b)
V-I Area
Sequence N
(c)
Power CV Difference
(d)
V-I Area Difference
(e)
Power CV Threshold
Power CV Difference Threshold
(f)
V-I Area Threshold
V-I Area Difference Threshold

Download: Download high-res image (1MB)

Download: Download full-size image

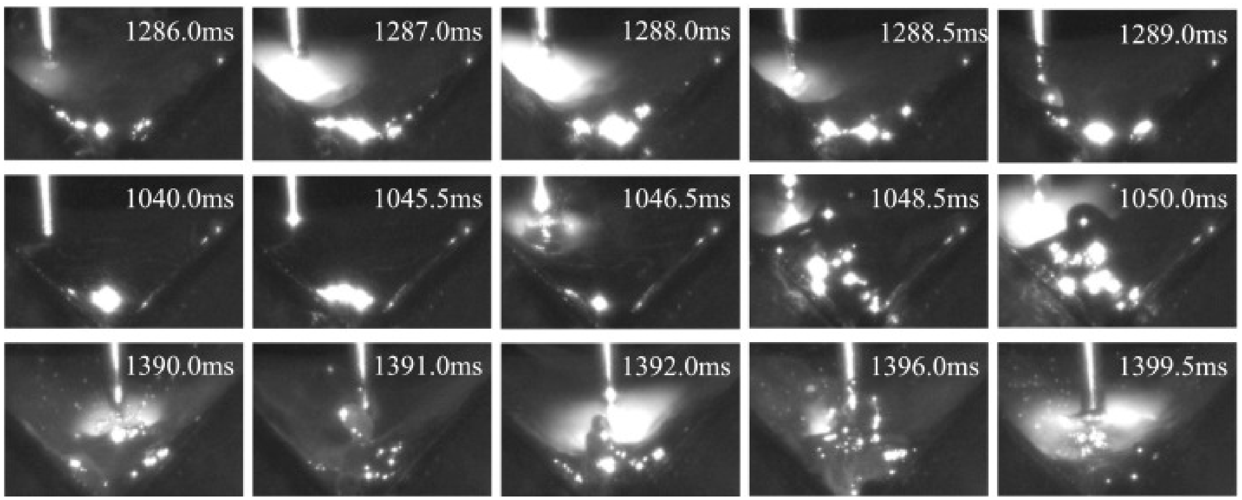
Fig. 6. Scatter plot of selected physical features: (a) Power CV; (b) V-I Area; (c) Power CV Difference; (d) V-I Area Difference; (e) the 2D classification diagram of Power CV and Power CV Difference; (f) the 2D classification diagram of V-I Area and V-I Area Difference.

To comprehensively represent arc dynamics, four representative features were selected: two sensitive to transient perturbations and two reflecting periodicity. Features integrating current and voltage, such as Power CV (Fig. 6(a)) and Power CV Difference (Fig. 6(c)), respond rapidly to short-lived instabilities and detect numerous unstable points. By contrast, periodicity-oriented features such as V—I Area and V—I Area Difference (Fig. 6(b), (d)) highlight deviations from the regular process rhythm, identifying fewer but distinct instability events associated with periodic arc behavior.

While individual features are informative, the 2D classification diagrams in Fig. 6(e) and (f) illustrate how expert-defined thresholds partition the feature space into four distinct quadrants. The bottom-left quadrant corresponds to stable states consistently identified by both features, whereas the top-right quadrant reflects a strong consensus on severe instability. The remaining two quadrants (top-left and bottom-right) are particularly revealing, as they contain points flagged as unstable by only one feature, thereby highlighting the complementary diagnostic perspectives captured by different physical features. This observation highlights that no single feature is sufficient for robust monitoring and provides compelling justification for adopting a multi-heuristic framework, such as Data Programming, to systematically integrate these diverse and complementary sources of information.

To visually substantiate the relationship between extracted features and actual arc phenomena, Fig. 7 presents high-speed camera frames corresponding to three representative data points explicitly marked in the Power CV time-series of Fig. 6(a). Point

N128 (1280–1290 ms) lies well below the instability threshold. The corresponding images reveal a complete droplet transfer cycle (1286–1289 ms) with stable arc attachment and smooth detachment, confirming stable behavior. By contrast, point N104 reflects a severe instability event characterized by arc extinction and subsequent re-ignition. Its Power CV value lies in the upper range, and the associated frames show re-ignition at 1046.5 ms, accompanied by irregular droplet morphology and intense pool oscillation. Similarly, point N139 (1390–1400 ms), also above the instability threshold, corresponds to dynamic variations in cathode height induced by welding torch moving. The images demonstrate unstable behavior, including irregular droplet formation and spatter ejection. These instabilities reflect common process deviations in industrial robotic welding. Collectively, these observations confirm that the extracted physical features provide robust, quantitative indicators of arc stability, with the high-speed images illustrating their direct correspondence to arc morphology and validating their diagnostic effectiveness.

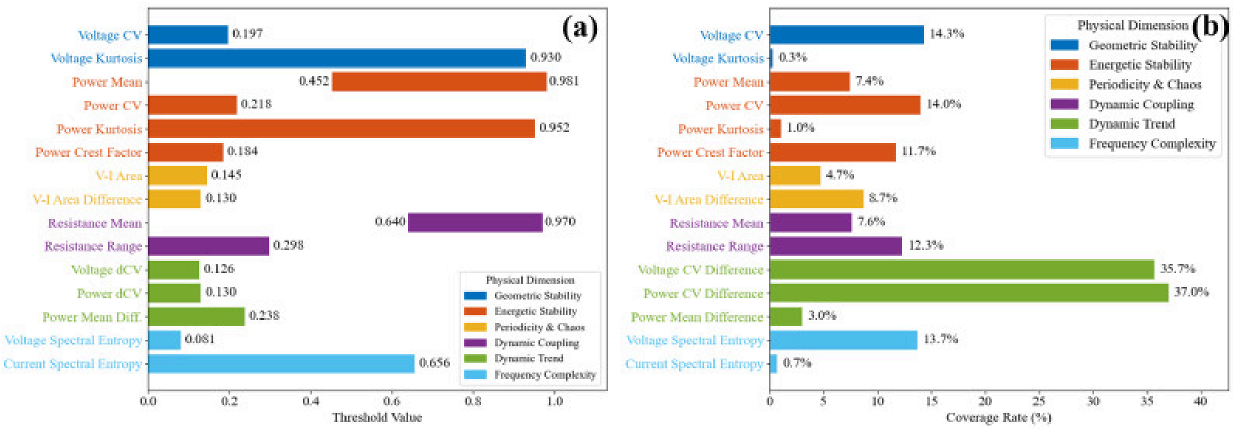


Download: Download high-res image (541KB)

Download: Download full-size image

Fig. 7. Visual correspondence between extracted physical features and arc phenomena based on high-speed camera frames.

Analysis of expert-defined thresholds, normalized to the [0, 1] range (Fig. 8(a)), highlights the diagnostic characteristics of individual features. Most instability thresholds lie below 0.3, indicating that modest increases in feature magnitude are sufficient to signal instability. This reflects their shared property of "deviation from low values as an anomaly." In contrast, Power Mean and Resistance Mean exhibit two-sided range thresholds, confirming that their stable operation is confined within a bounded interval, with deviations in either direction implying instability.



Download: Download high-res image (462KB)

Download: Download full-size image

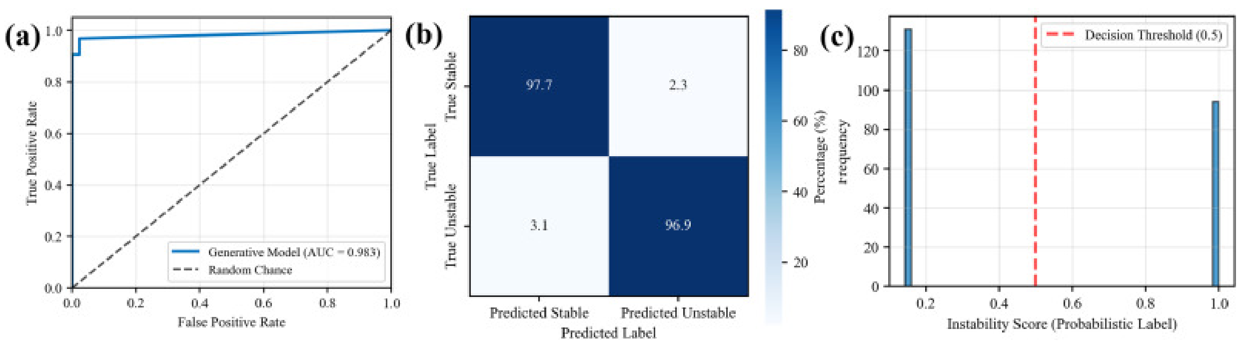
Fig. 8. Determination of physical feature threshold: (a) The thresholds of physical features for arc stability identification; (b) the diagnostic coverage rate of each physical feature.

Notably, the thresholds for Voltage Kurtosis and Power Kurtosis exceed 0.9, indicating that only extreme signal impulses activate their diagnostic rules. The diagnostic coverage of each feature when used independently (Fig. 8(b)) further quantifies this behavior. Higher-order dynamic features, such as Voltage CV Difference and Power CV Difference, show the greatest sensitivity by capturing the evolving trends of arc morphology beyond its instantaneous state. In contrast, Voltage Kurtosis, Power Kurtosis, and Current Spectral Entropy identify less than 1% of events, consistent with their extremely high thresholds.

This low coverage indicates that, under a single-threshold criterion, these features exhibit limited diagnostic capacity for capturing the full spectrum of instability events.

5.2. Performance of the generative model and label quality assessment

This section evaluates the effectiveness of the proposed Physics-Guided Data Programming approach by examining the performance of the Generative Model. The binary predictions of the Generative Model were obtained by thresholding its probabilistic outputs at 0.5. On an independent ground-truth labels set, the model achieved an accuracy of 97.3%, a Macro F1-Score of 0.969, and an AUC of 0.983 (Fig. 9(a)). Its confusion matrix (Fig. 9(b)) further corroborates the model's reliability, showing consistently high precision and recall for the UNSTABLE class with only limited misclassifications.



Download: Download high-res image (252KB)

Download: Download full-size image

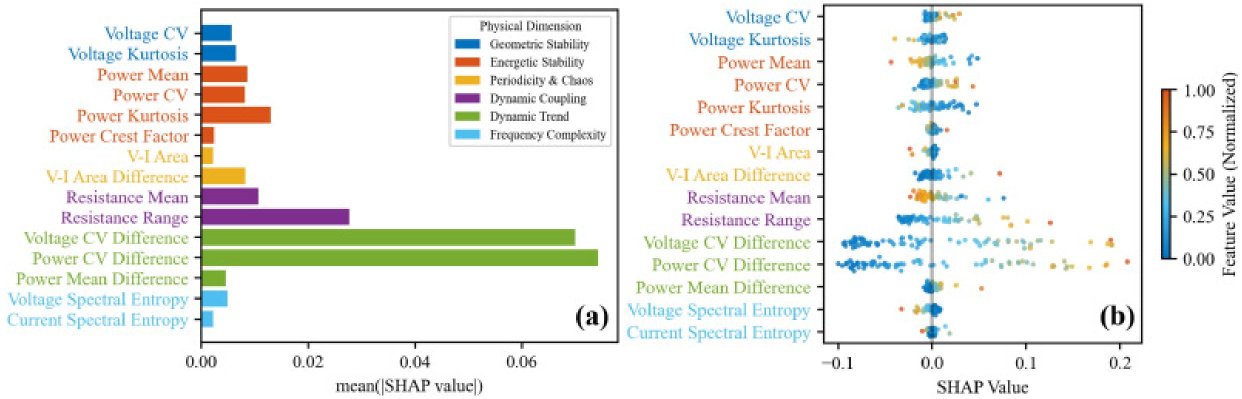
Fig. 9. Efficacy of the generative models: (a) ROC curves; (b) confusion matrix of generative model; (c) probabilistic label.

In addition to its classification performance, Fig. 9(c) illustrates two discrete bars at approximately 0.17 (stable) and 0.98 (unstable), with virtually no samples in the intermediate range. This outcome indicates the strong discriminative power of the physics-guided Labeling Functions (LFs). Because these expert-defined heuristics were designed to be mutually exclusive, the Generative Model was able to integrate the weak supervision signals into high-confidence assignments for the majority of samples.

Collectively, these results demonstrate that the Generative Model achieves high accuracy on the independent validation set and simultaneously provides reliable, high-confidence labels for large-scale training data. This confirms that the Physics-Guided Data Programming approach effectively consolidates multiple heuristic rules into consistent labels, thereby substantially reducing the dependence on manual annotation.

5.3. Interpretability and physical mechanism analysis of the downstream model

To move beyond opaque “black-box” predictions and gain deeper insights into the physical mechanisms driving the model's decisions, SHAP is leveraged to elucidate the model's internal reasoning. This analysis not only enhances model transparency but also elucidates the structured reasoning embedded in the data-driven model. Here, the analysis relies on PGDP-15, a downstream MLP classifier trained with 15 physics-guided features and labels synthesized by the generative model, enabling a direct evaluation of the contribution of physics-derived inputs. The global feature importance, shown in Fig. 10(a), is determined by the mean absolute SHAP value for each feature. A clear hierarchy emerges: the dynamic trend features, Power CV Difference and Voltage CV Difference, are identified as the most influential predictors by a significant margin, confirming that the rate of change in process fluctuations is the dominant determinant of arc instability. Fig. 10(b) provides a finer-grained view of these contributions at the sample level, illustrating that elevated CV Difference values consistently shift predictions toward the UNSTABLE class.



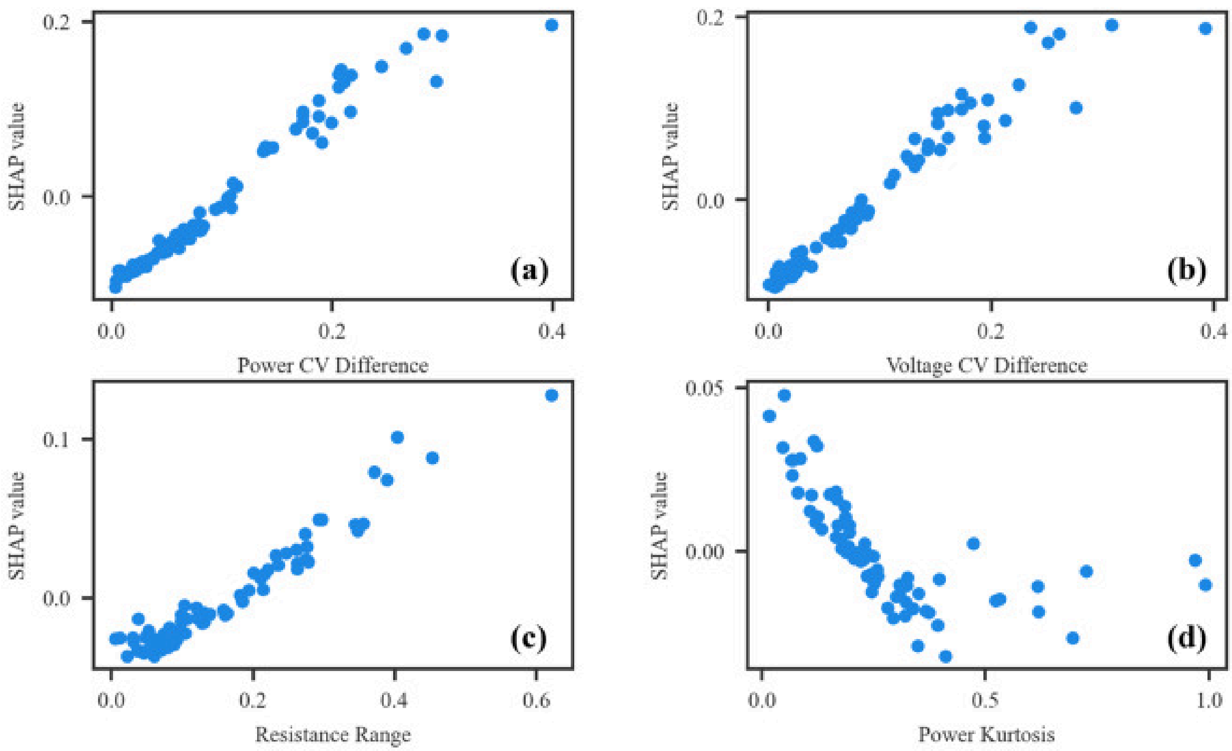
Download: Download high-res image (554KB)

Download: Download full-size image

Fig. 10. Importance of physical features in arc instability detection evaluated by SHAP analysis: (a) overall importance of the features; (b) feature contribution of individual samples.

Despite Power Kurtosis's relatively high ranking in the SHAP importance plot, it demonstrated poor diagnostic coverage (1%) in threshold screening. This indicates that Power Kurtosis is not an effective standalone indicator but functions as a synergistic feature through non-linear interactions with other inputs. Such relationships cannot be captured by simple threshold-based heuristics but are effectively leveraged by the data-driven model. This underscores the principal advantage of our approach in identifying complex dependencies beyond conventional rule-based logic.

The SHAP dependence plots in Fig. 11 reveal clear and physically consistent relationships between the four most influential features and arc stability. Power CV Difference, Voltage CV Difference, and Resistance Range exhibit monotonic positive correlations with SHAP values, indicating that increased signal fluctuations are strongly associated with arc length oscillations and irregular droplet detachment, which disrupt the thermal equilibrium of the molten pool. In contrast, Power Kurtosis shows a distinct negative correlation, where lower kurtosis corresponds to higher instability. Under stable spray transfer, the power distribution is highly concentrated, yielding high kurtosis. Irregular arc behavior disperses the energy distribution, reduces kurtosis, and broadens the signal profile, making kurtosis a sensitive marker of instability. This alignment between SHAP attributions and welding physics validates that the downstream model has effectively internalized domain-specific knowledge from the weak supervision signals. Importantly, identifying this high-order statistical indicator highlights the framework's ability to go beyond conventional first-order thresholds and uncover physically meaningful yet non-intuitive features that are difficult to encode through manual expert rules.



Download: Download high-res image (164KB)

Download: Download full-size image

Fig. 11. The variation of SHAP value with the feature values: (a) Power CV Difference; (b) Voltage CV Difference; (c) Resistance Range; (d) Power Kurtosis.

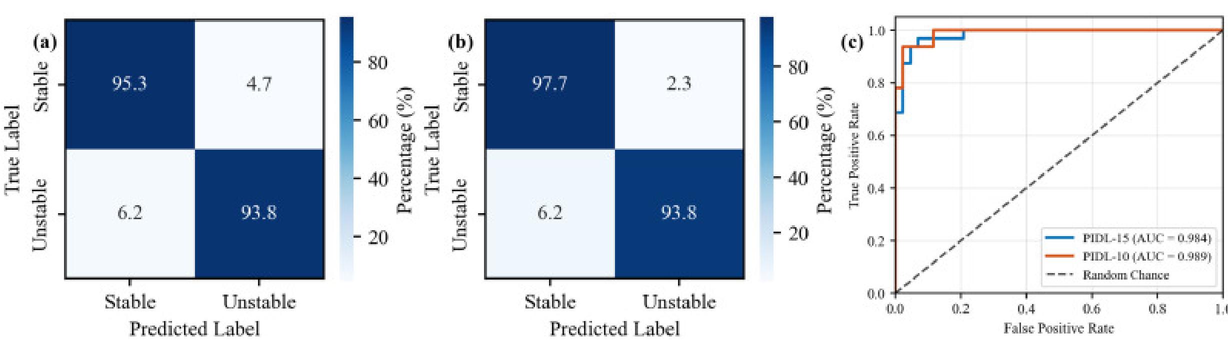
5.4. Performance optimization and validation of downstream models

The initial PGDP-15 model integrated a comprehensive set of 15 physical features. To optimize model complexity without compromising physical fidelity, a SHAP-guided pruning strategy was employed. Analysis indicated that the top-10 ranked features account for over 95% of the total global SHAP attribution value. Consequently, the five lowest-ranked predictors, including Power Crest Factor, V—I Area, Power Mean Difference, Voltage Spectral Entropy, and Current Spectral Entropy, were excluded. Crucially, a physical sufficiency analysis confirms that this reduction represents the elimination of redundancy rather than a loss of domain information. Specifically, the mechanism of process chaos is more robustly captured in the time domain by the V—I Area Difference, which offers superior temporal resolution for detecting transient droplet transfer anomalies compared to the global statistics of spectral entropy. Similarly, static magnitude indicators were found to be subsumed by their dynamic counterparts (e.g., Power CV Difference), which exhibit higher sensitivity to the precursors of instability. An identical MLP architecture was then retrained on this physically optimized feature set.

Both PGDP-15 and PGDP-10 were evaluated against generated labels for internal consistency and against ground-truth labels for generalization (Table 5). The optimized PGDP-10 model not only preserved perfect recall for unstable states but also improved accuracy and F1-scores over PGDP-15 across both evaluation settings. The gains were consistent, with accuracy rising from 0.964 to 0.973 on generated labels and from 0.947 to 0.960 on ground-truth labels, while F1-scores improved from 0.959 to 0.969 and from 0.938 to 0.952, respectively. Crucially, the maintenance of perfect recall (1.000) for unstable states serves as empirical validation that the feature pruning process did not discard predictors critical for detecting subtle instability modes. Simultaneously, the improved precision indicates a reduction in false positives for stable states. Confusion matrices (Fig. 12(a), (b)) corroborate these improvements, and ROC curves (Fig. 12(c)) show uniformly high AUC values across all states, underscoring the strong discriminative capability of the optimized feature set.

Table 5. Performance comparison for PGDP-15 and PGDP-10.

Metric	PGDP-15 (generated labels)	PGDP-15 (ground-truth labels)	PGDP-10 (generated labels)	PGDP-10 (ground-truth labels)
Accuracy	0.964	0.947	0.973	0.960
Precision	0.922	0.938	0.940	0.968
Recall	1.000	0.938	1.000	0.938
F1-Score	0.959	0.938	0.969	0.952
AUC	0.999	0.984	1.000	0.989



Download: Download high-res image (258KB)

Download: Download full-size image

Fig. 12. Evaluation of PGDP models on ground-truth labels: (a) Confusion matrices for PGDP-15; (b) confusion matrices for PGDP-10; (c) ROC curves (AUC) for PGDP-15 and PGDP-10.

Overall, SHAP-guided optimization distilled the most informative physical features, yielding a more parsimonious model with superior stability and generalization. These results demonstrate that weakly supervised models trained with generated labels can achieve high fidelity to physical rules while reducing reliance on costly manual annotation.

5.5. Comparative analysis with baseline methods

The quantitative performance of the proposed PGDP-10 framework was benchmarked against seven baseline methods summarized in Table 6. As shown by the results, PGDP-10 achieves the highest F1-score (0.952), outperforming statistical, unsupervised, and traditional supervised approaches. Statistical monitoring methods, including the Control Chart and PCA, exhibit high precision but substantially lower recall (0.438–0.531), indicating that rigid univariate thresholds and linear reconstruction-based criteria are insufficient to capture subtle, non-linear precursors of arc instability. This limitation is particularly pronounced in dynamic scenarios such as weaving welding, where the intrinsic periodic fluctuations of the signal baseline violate the stationarity assumption of traditional Control Charts. Consequently, static thresholds struggle to distinguish between benign weaving motions and genuine defect precursors.

Table 6. Quantitative performance comparison of the proposed PGDP-10 framework against baseline methods.

Methods	**Accuracy**	**Precision**	**Recall**	**F1-Score**
3-Sigma Control Chart	0.746	0.933	0.437	0.595
PCA	0.786	0.944	0.531	0.680
Isolation Forest	0.733	0.650	0.812	0.722
LOF	0.733	0.676	0.718	0.696
One-Class SVM	0.733	0.620	0.968	0.756
Logistic Regression	0.906	0.962	0.812	0.881
Random Forest	0.933	0.965	0.875	0.918

Methods	Accuracy	Precision	Recall	F1-Score
PGDP-10	0.960	0.968	0.938	0.952

The semi-supervised One-Class SVM attains the highest recall but at the expense of markedly reduced precision (0.620), suggesting a tendency to over-identify benign process variations, such as inherent fluctuations induced by torch weaving, as instability events. Similarly, unsupervised density- and isolation-based methods (iForest and LOF) yield moderate F1-scores. This performance deficit highlights the inherent limitation of relying solely on feature-space geometry when dealing with unbalanced datasets containing partially unknown anomaly classes.

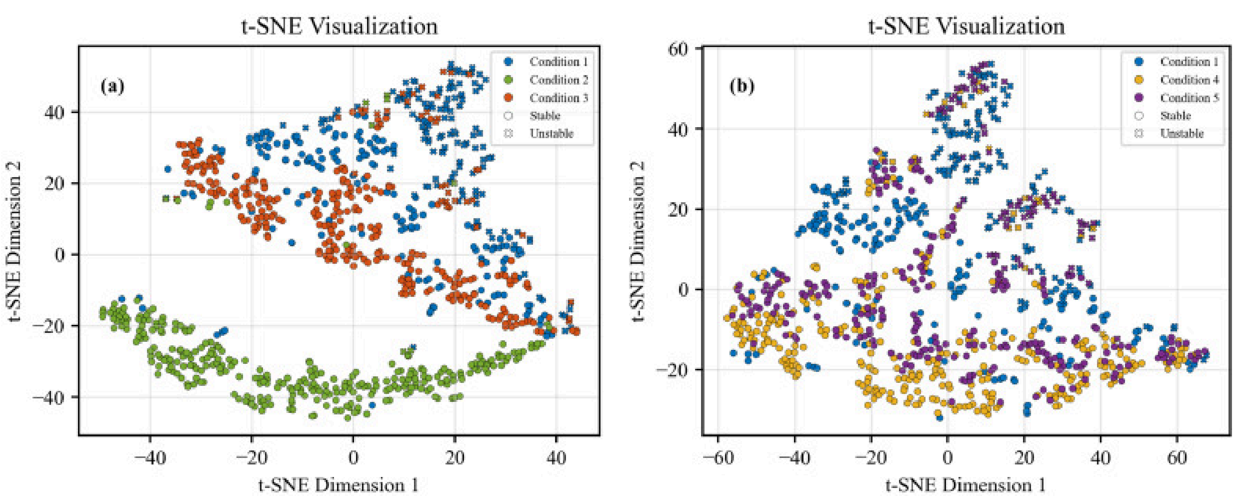
In the comparison with supervised classifiers trained using identical probabilistic weak labels, Logistic Regression achieves a lower F1-score (0.881) than both Random Forest and PGDP-10, indicating that linear decision boundaries are insufficient to fully characterize the complex relationships between physics-informed features and arc stability. While Random Forest serves as a strong baseline, PGDP-10 surpasses it by 3.4% in F1-score and achieves a notably higher recall (0.938 compared to 0.875). Crucially, the superior performance of PGDP-10 over all baseline categories highlights the unique advantage of the Data Programming paradigm. Unlike unsupervised or semi-supervised methods that struggle with feature space ambiguity in dynamic weaving processes, and distinct from rigid statistical thresholds that propagate label noise, PGDP-10 effectively integrates probabilistic denoising with deep representation learning. This allows the discriminative model to learn generalized, high-dimensional instability patterns that extend beyond the limited coverage of individual expert heuristics, thereby achieving the most favorable balance between precision and recall for industrial monitoring.

5.6. Generalizability study in different welding conditions

To further validate the effectiveness and robustness of the proposed framework, the pretrained PGDP-10 model was applied to arc stability prediction under Conditions 2–5.

Notably, the arc stability labels for these conditions were not directly observed but instead generated by the proposed label-generation model. Their reliability will be further examined through high-speed imaging of unstable events. By eliminating the need for extensive manual annotation across varied operating parameters, this workflow demonstrates significant scalability, enabling the rapid deployment of high-fidelity monitoring systems in diverse industrial welding conditions.

The two-dimensional feature distributions of the generated labels are visualized using t-SNE in Fig. 13, divided into two comparative groups: Conditions 1–3 (Fig. 13(a)) and Conditions 1, 4, and 5 (Fig. 13(b)). In both projections, a distinct topological structure is observed where stable samples from all diverse conditions cluster predominantly on one side of the feature space, while unstable samples cluster on the opposite side. This coherent separation indicates that the five welding scenarios, despite their parametric differences, share a common physical domain without evident domain shift [31]. This consistency ensures that the extracted features and the trained PGDP model remain robust across expanded operating windows, supporting reliable generalization.



Download: Download high-res image (818KB)

Download: Download full-size image

Fig. 13. The t-SNE of generated labels: (a) Conditions 1, 2 and 3; (b) Conditions 1, 4 and 5.

Furthermore, the relative geometric positions of the data points quantify the degree of process stability: the greater the Euclidean distance from the unstable cluster center, the higher the inherent arc stability. As illustrated in Fig. 13(a), the distribution aligns with the arc signals in Fig. 2, showing that Condition 2, characterized by optimally tuned parameters that minimize arc-length fluctuations, is situated furthest from the unstable region, reflecting higher overall stability than both the frequently unstable Condition 1 and the moderately unstable Condition 3. A similar spatial stratification is evident in Fig. 13(b), where Conditions 4 and 5 also exhibit clear separability from the unstable regime of Condition 1, further confirming the framework's capability to discriminate stability states across a broad spectrum of welding conditions.

The generalization performance of PGDP-10 across four distinct unseen operating conditions (Conditions 2–5) is quantitatively assessed in Table 7. The model demonstrates robust adaptability, maintaining an accuracy consistently above 96.6% across all scenarios. Critically, the model achieves a perfect Recall of 1.000 in Conditions 2, 4, and 5, and a high Recall of 0.944 in Condition 3. This indicates that the physics-informed framework successfully captures nearly every instability event regardless of the operating window, satisfying the stringent “zero-defect” requirement of industrial quality monitoring. In terms of the F1-score, a divergence in performance is observed based on the nature of the instability. Conditions 2 and 5 exhibit superior performance, with F1-scores of 0.941 and 0.939, respectively. Notably, Condition 5 maintains this high fidelity despite a high frequency of instability (69 counts), demonstrating that the model's robustness is not compromised by the density of defect occurrence. Fundamentally, this sustained performance on unseen datasets serves as empirical validation that the model has learned robust, transferable physical laws governing arc stability.

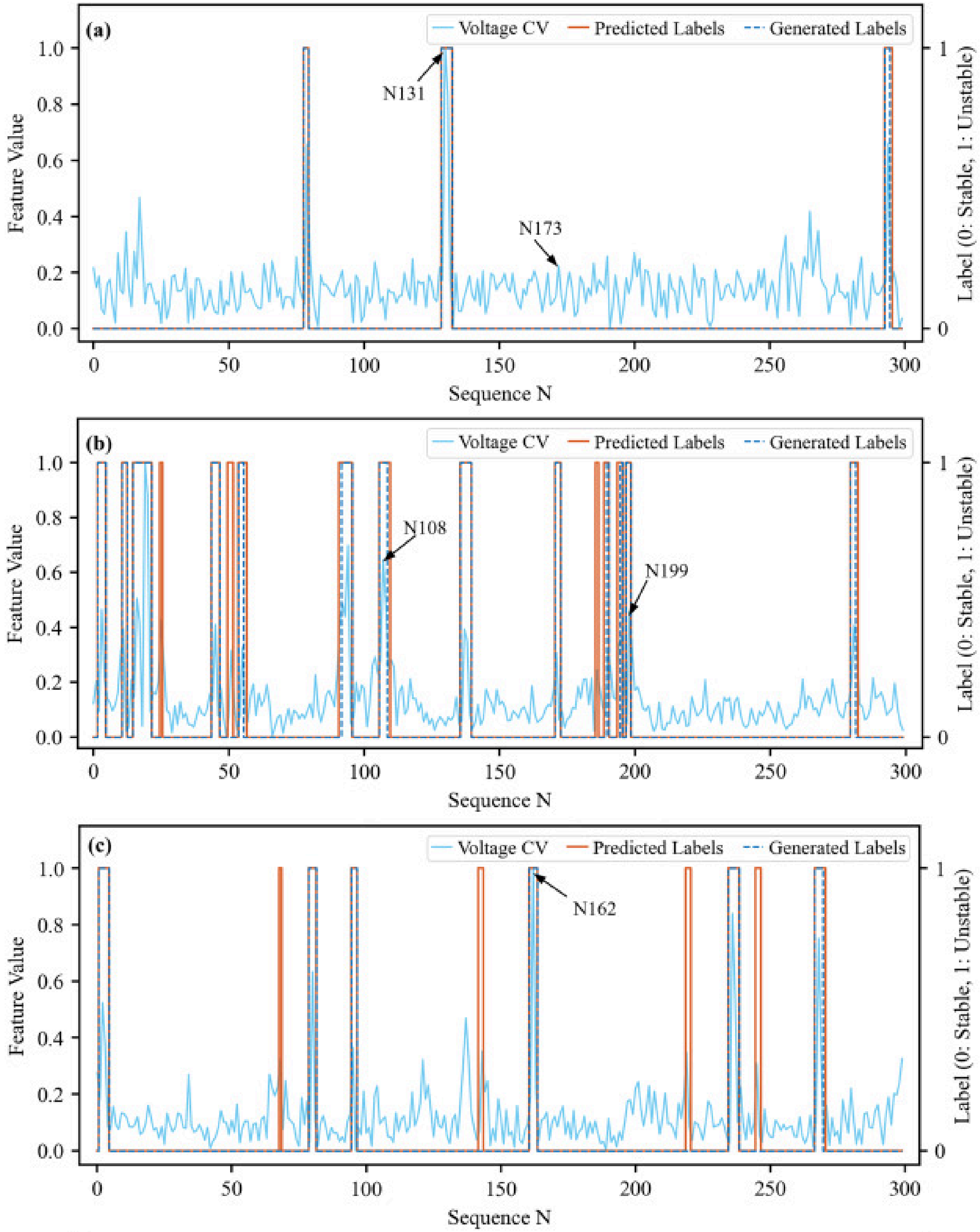
Table 7. Generalization performance across conditions.

	Accuracy	**Precision**	**Recall**	**F1-Score**	**Unstable count**
Condition 2	0.996	0.889	1.000	0.941	8

	Accuracy	Precision	Recall	F1-Score	Unstable count
Condition 3	0.966	0.809	0.944	0.872	36
Condition 4	0.983	0.792	1.000	0.884	19
Condition 5	0.970	0.885	1.000	0.939	69

Conversely, a divergence in F1-scores is observed in Conditions 3 and 4, where values decrease to 0.872 and 0.884, respectively. Corroborated by the t-SNE topological analysis (Fig. 13), this variation is attributed to distinct instability mechanisms rather than a fundamental domain shift. While the training set (Condition 1) and Condition 5 are characterized by 'gross instability' events (e.g., complete arc extinction or stubbing) that form topologically distinct clusters, Conditions 3 and 4 represent 'transitional' or 'meta-stable' regimes. The instability in these scenarios manifests as subtle stochastic disturbances, such as minor pool oscillations or micro-jitters in arc length, which lie proximal to the model's decision boundary in the feature space. Consequently, the slightly lower Precision (0.809 and 0.792) reflects the model's conservative detection strategy: it prioritizes capturing these ambiguous, near-boundary precursors to ensure high Recall. In the context of robotic welding, this high sensitivity is structurally advantageous, as identifying potential instability precursors aligns with the stringent 'zero-defect' protocol, where missing a defect (False Negative) is far more detrimental than a conservative false alarm.

The time-series prediction results across the five operating conditions are illustrated in Fig. 14(a)–(d), where the Voltage CV is plotted alongside predicted labels to elucidate physical interpretability. Consistent across all conditions, the predicted unstable regions align closely with peaks in Voltage CV, confirming that PGDP-10 effectively captures the physical signatures of arc instability. Notably, in instances such as Fig. 14(a) and (c), a small subset of samples with elevated Voltage CV are classified as unstable by the model even when their generated labels remain stable. This sensitivity to high feature values suggests that PGDP-10 functions as an early-warning indicator for potential instability—a conservative behavior desirable in industrial monitoring, where missed detections are significantly more detrimental than occasional false alarms.


(a)
Voltage CV
Predicted Labels
Generated Labels
N131
N173
Feature Value
Label (0: Stable, 1: Unstable)
Sequence N
(b)
Voltage CV
Predicted Labels
Generated Labels
N108
N199
Feature Value
Label (0: Stable, 1: Unstable)
Sequence N
(c)
Voltage CV
Predicted Labels
Generated Labels
N162
Feature Value
Label (0: Stable, 1: Unstable)
Sequence N

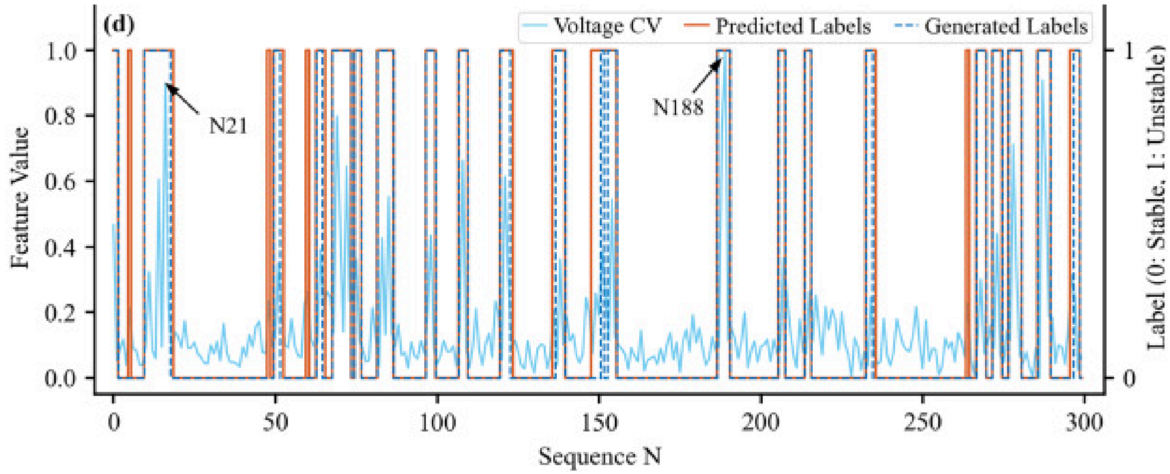
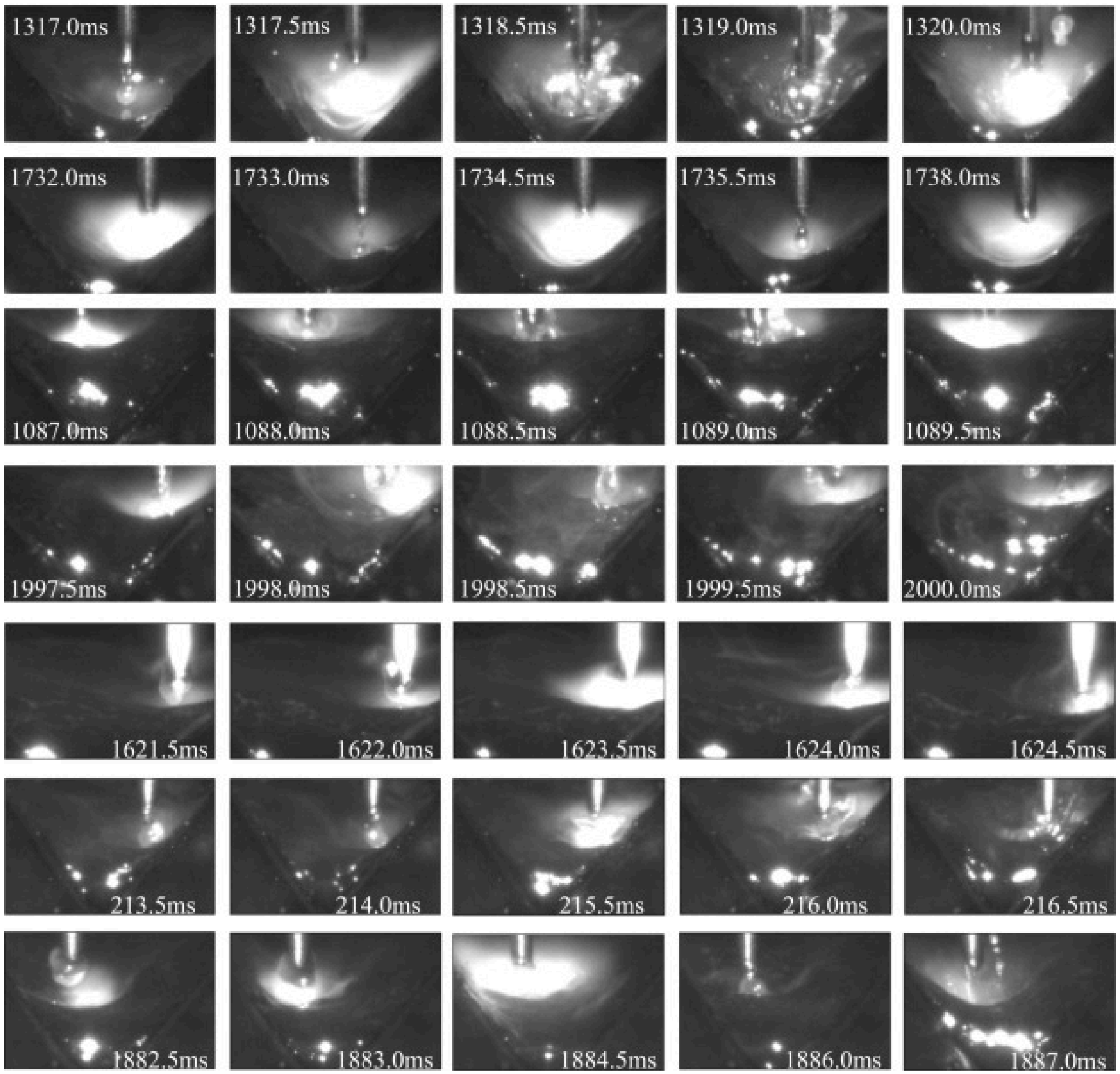


Fig. 14. PGDP-10 model predictions of arc stability under different welding conditions: (a) Condition 2; (b) Condition 3; (c) Condition 4; (d) Condition 5.

To further validate the generative labels and the predictive fidelity of PGDP-10, representative points from Conditions 2 through 5 were selected from Fig. 14 for cross-referencing with synchronized high-speed imaging (Fig. 15), revealing a distinct correspondence between captured physical phenomena and model-predicted events. In Condition 2, point N131 (1310–1320 ms) reveals a wire explosion event. Although such anomalies are rare under this optimized condition, this specific instability is attributed to the coupling effect between the torch weaving motion and the hydrodynamic behavior of the molten pool. As the arc traverses the dynamic, undulating liquid surface, instantaneous variations in the cathode boundary height occur. These fluctuations momentarily disrupt the arc length stability, triggering an explosive detachment before the power source self-regulation can compensate. By contrast, Point N173 (1730–1740 ms), selected from a torch weaving segment, remained fully stable. This point demonstrates that, despite perturbations introduced by weaving, maintaining the cathode height within an optimal range ensures stability, providing direct evidence that arc instability stems primarily from excessive arc-length fluctuations rather than the weaving motion itself. In Condition 3, the arc behavior differs substantially. Prolonged dwell during weaving causes the molten pool surface to rise, elevating the arc root position and triggering unstable phenomena. This is confirmed at point N108 (1080–1090 ms) and point N199 (1990–2000 ms), where irregular droplet shapes and spattering are clearly observed. Condition 4 exhibits irregular droplet morphology disrupting metal transfer (Point N162, 1620–1630 ms).



Download: Download high-res image (1012KB)

Download: Download full-size image

Fig. 15. High-speed welding arc images of representative time sequence points in Condition 2 to Condition 5.

Finally, Condition 5 highlights complex dynamic instabilities, including severe molten pool oscillation induced by arc fluctuations (Point N21, 210–220 ms) and explosive spatter caused by delayed droplet detachment (Point N188, 1880–1890 ms). These results provide compelling evidence that the framework accurately forecasts arc instability driven by diverse physical mechanisms across different operating conditions. Furthermore, as evidenced by the synchronized sequences, mapping these temporal detections to the deterministic torch trajectory allows for the precise identification of defect spatial regions, laying the essential foundation for spatial defect prediction and subsequent targeted process optimization.

6. Conclusion

This study proposed a Physics-Guided Data Programming framework for arc stability monitoring in GMAW. The main conclusions are as follows:

(1) Scalable weak supervision for label generation. By incorporating Data Programming, expert-defined physical heuristics were systematically transformed into labeling functions, enabling large-scale label generation while reducing the high cost of manual annotation. The effectiveness of this generative labeling process was rigorously validated on an independent gold-standard dataset, where the Generative Model achieved an accuracy of 97.3%, a Macro F1-score of 0.969, and an AUC of 0.983. Furthermore, the successful application of this labeling pipeline to unseen welding conditions (Conditions 2–5) without additional manual effort demonstrates its strong scalability for diverse industrial scenarios. These results demonstrate that the proposed weak supervision strategy not only alleviates the annotation bottleneck but also yields trustworthy labels for subsequent model training.

(2) Physics-driven interpretability and insight. The framework encodes physical knowledge from six fundamental dimensions to derive 15 representative features. SHAP-based attribution went beyond simple feature selection to

verify the physical consistency of the learned patterns. It confirmed Voltage and Power CV differences as dominant indicators while crucially identifying Power Kurtosis as a high-order precursor of instability. This finding demonstrates the model's capacity to capture complex distributional shifts that lie beyond the scope of conventional threshold-based heuristics, surpassing the detection capabilities of static expert rules.

(3) Superior performance and generalization. The optimized PGDP-10 model achieved an F1-score of 0.952 under benchmark conditions, significantly surpassing traditional statistical baselines. Furthermore, the model exhibited robust generalization across four unseen operating conditions (Conditions 2–5), maintaining accuracies consistently above 0.966. These results validate the adaptability of the framework to diverse process variations.

The proposed PGDP framework lays a solid foundation for physics-guided weak supervision in welding process monitoring. Nevertheless, the present study is limited to passive, open-loop monitoring, and the weak supervision rules are currently tailored to the GMAW process. Future work will extend this framework toward spatial defect prediction by linking temporal anomaly patterns with weld geometry, and further embedding such predictions into closed-loop control strategies for real-time process regulation. In parallel, computational efficiency will be further optimized to enable practical deployment on resource-constrained embedded platforms. These advancements are expected to substantially enhance the robustness, scalability, and industrial applicability of intelligent welding quality control systems.

CRediT authorship contribution statement

Lei Zhang: Writing – original draft, Methodology, Data curation, Conceptualization. **Lin Chen:** Methodology, Funding acquisition, Data curation. **Binqi Jia:** Data curation.

Sichuang Yang: Writing – review & editing, Conceptualization. **Minling Pan:** Writing – review & editing. **Haihong Pan:** Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 51465005); by the Guangxi Science and Technology Major Program (Grant Nos. AA18118002 and 2025JFJF27003); by the Guangxi Natural Science Foundation (Grant No. 2025JJH170094, 2026GXNSFHA00640304 and 2026GXNSFHA00640064); and by the Innovation Project of Guangxi Graduate Education (Grant Nos. YCBZ2024015 and A3010022002).

Data availability

Data will be made available on request.

References

[1] Y. Ma, J. Fan, J. Hou, Y. Fu, R. Tao, S. Wang, *et al.*
Vision-based autonomous robotic arc welding: state-of-the-art review and perspectives
IEEE Trans Autom Sci Eng, 22 (2025), pp. 21651-21673
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[2] H.H. König, N.H. Pettersson, A. Durga, S. Van Petegem, D. Grolimund, A.C. Chuang, *et al.*

Solidification modes during additive manufacturing of steel revealed by high-speed X-ray diffraction
Acta Mater, 246 (2023), Article 118713
View PDF View article View in Scopus ↗ Google Scholar ↗

[3] Z. Han, S. Li, X. Chen, B. Huang, J. Sun, Q. Zhang, *et al.*
DFW-YOLO: YOLOv5-based algorithm using phased array ultrasonic testing for weld defect recognition
Nondestruct Test Eval, 40 (6) (2025), pp. 2516-2539
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[4] D. Kumar, S. Jha
Deposition strategy correlation with texture development, geometric homogeneity, and mechanical anisotropy in robotic WAAM-fabricated aluminum alloy thick walls
J Manuf Process, 145 (2025), pp. 545-555
View PDF View article Google Scholar ↗

[5] G. Mattera, S.P. Chozaki, J. Norrish
Advances in machine learning for parameters optimisation and in-situ monitoring of wire arc additive manufacturing
Weld World (2025), pp. 1-30
Google Scholar ↗

[6] T. Hauser, R.T. Reisch, P.P. Breese, B.S. Lutz, M. Pantano, Y. Nalam, *et al.*
Porosity in wire arc additive manufacturing of aluminium alloys
Addit Manuf, 41 (2021), Article 101993
View PDF View article View in Scopus ↗ Google Scholar ↗

[7] S. Han, G. Liu, X. Tang, L. Xu, H. Cui, C. Shao

Effect of molten pool behaviors on welding defects in tandem NG-GMAW based on CFD simulation
Int J Heat Mass Transfer, 195 (2022), Article 123165
View PDF View article View in Scopus ↗ Google Scholar ↗

[8] G. Liang, G. Qin, X. Fan, J. Wang
In-depth analysis of heat transfer and fluid dynamic behaviors of molten pool under pulsed characteristics in cold metal transfer welding via numerical simulation
Int J Heat Mass Transfer, 251 (2025), Article 127405
View PDF View article View in Scopus ↗ Google Scholar ↗

[9] A.S. Madhvacharyula, A.V.S. Pavan, S. Gorthi, S. Chitral, N. Venkaiah, D.V. Kiran
In situ detection of welding defects: a review
Weld World, 66 (4) (2022), pp. 611-628
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[10] W. Li, H. Zhang, G. Wang, G. Xiong, M. Zhao, G. Li, *et al.*
Deep learning based online metallic surface defect detection method for wire and arc additive manufacturing
Robot Comput Integr Manuf, 80 (2023), Article 102470
View PDF View article View in Scopus ↗ Google Scholar ↗

[11] J.Y. Alcaraz, A. Sharma, T. Tjahjowidodo
Predicting porosity in wire arc additive manufacturing (WAAM) using wavelet scattering networks and sparse principal component analysis
Weld World, 68 (4) (2024), pp. 843-853
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[12] H. Mu, F. He, L. Yuan, P. Commins, J. Xu, Z. Pan

High-frequency real-time bead geometry measurement in wire arc additive manufacturing based on welding signals
IEEE Trans Ind Inform, 21 (3) (2024), pp. 2630-2639
Google Scholar ↗

[13] Z. Wang, L. Li, H. Chen, X. Wu, Y. Dong, J. Tian, Q. Zhang
Penetration recognition based on machine learning in arc welding: a review
Int J Adv Manuf Technol, 125 (9) (2023), pp. 3899-3923
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[14] R. Li, H. Ma, R. Wang, H. Song, X. Zhou, L. Wang, *et al.*
Application of unsupervised learning methods based on video data for real-time anomaly detection in wire arc additive manufacturing
J Manuf Process, 143 (2025), pp. 37-55
View PDF View article Google Scholar ↗

[15] H. Song, C. Li, Y. Fu, R. Li, H. Zhang, G. Wang
A two-stage unsupervised approach for surface anomaly detection in wire and arc additive manufacturing
Comput Ind, 151 (2023), Article 103994
View PDF View article View in Scopus ↗ Google Scholar ↗

[16] M.A. Rahman, S. Jamal, M.V. Cruz, B. Silwal, H. Taheri
In situ process monitoring of multi-layer deposition in wire arc additive manufacturing (WAAM) process with acoustic data analysis and machine learning
Int J Adv Manuf Technol, 132 (9) (2024), pp. 5087-5101
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[17] G. Mattera, J. Polden, J. Norrish

Monitoring the gas metal arc additive manufacturing process using unsupervised machine learning
Weld World, 68 (11) (2024), pp. 2853-2867
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[18] G. Mattera, M. Vozza, J. Polden, L. Nele, Z. Pan
Frequency informed convolutional autoencoder for in situ anomaly detection in wire arc additive manufacturing
J Intell Manuf, 36 (8) (2025), pp. 5819-5834
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[19] Y. Kim, T. Lee, Y. Hyun, E. Coatanea, S. Mika, J. Mo, *et al.*
Self-supervised representation learning anomaly detection methodology based on boosting algorithms enhanced by data augmentation using StyleGAN for manufacturing imbalanced data
Comput Ind, 153 (2023), Article 104024
View PDF View article View in Scopus ↗ Google Scholar ↗

[20] G. Mattera, J. Polden, A. Caggiano, L. Nele, Z. Pan, J. Norrish
Semi-supervised learning for real-time anomaly detection in pulsed transfer wire arc additive manufacturing
J Manuf Process, 128 (2024), pp. 84-97
View PDF View article View in Scopus ↗ Google Scholar ↗

[21] B. Xue, D. Du, G. Peng, Y. Zhang, R. Li, Z. Li
Visual monitoring of weld penetration in aluminum alloy GTAW based on deep transfer learning enhanced by task-specific pre-training and semi-supervised learning
J Manuf Process, 133 (2025), pp. 1038-1050
View PDF View article View in Scopus ↗ Google Scholar ↗

[22] J. Zhang, C.Y. Hsieh, Y. Yu, C. Zhang, A. Ratner
A survey on programmatic weak supervision. arXiv preprint arXiv:2202.05433
(2022)
Google Scholar ↗

[23] A. Ratner, S.H. Bach, H. Ehrenberg, J. Fries, S. Wu, C. Ré
Snorkel: rapid training data creation with weak supervision
Proceedings of the VLDB endowment. International conference on very large data bases, vol. 11(3) (November 2017), p. 269
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[24] K. Meyer, V. Mahalec
Anomaly detection methods for infrequent failures in resistive steel welding
J Manuf Process, 75 (2022), pp. 497-513
View PDF View article View in Scopus ↗ Google Scholar ↗

[25] G. Wei, K. Hu, S. Chen, M. Yan
Experiment and simulation investigation of multiple cracks evolution at the weld toe
Int J Fatigue, 144 (2021), Article 106037
View PDF View article View in Scopus ↗ Google Scholar ↗

[26] J. Wang, Q. Sun, Y. Jiang, T. Zhang, J. Ma, J. Feng
Analysis and improvement of underwater wet welding process stability with static mechanical constraint support
J Manuf Process, 34 (2018), pp. 238-250
View PDF View article Google Scholar ↗

[27] Y. Huang, D. Yang, K. Wang, L. Wang, Q. Zhou

Stability analysis of GMAW based on multi-scale entropy and genetic optimized support vector machine
Measurement, 151 (2020), Article 107282
View PDF View article View in Scopus ↗ Google Scholar ↗

[28] S.M. Lundberg, S.I. Lee
A unified approach to interpreting model predictions
Adv Neural Inf Proces Syst, 30 (2017)
Google Scholar ↗

[29] G. Mattera, L. Nele
Machine learning approaches for real-time process anomaly detection in wire arc additive manufacturing
Int J Adv Manuf Technol (2025), pp. 1-26
Google Scholar ↗

[30] A.A. Melakhsou, M. Batton-Hubert, N. Casoetto
Welding fault detection and diagnosis using one-class SVM with distance substitution kernels and random convolutional kernel transform
Int J Adv Manuf Technol, 128 (1) (2023), pp. 459-477
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[31] E. Tzeng, J. Hoffman, K. Saenko, T. Darrell
Adversarial discriminative domain adaptation
Proceedings of the IEEE conference on computer vision and pattern recognition (2017), pp. 7167-7176
Google Scholar ↗

1 These authors contributed equally to this work.

View Abstract

Recommended articles

Based on reading popularity

Research article Full text access

Numerical and experimental investigation of electrode configuration effects on EHD heat transfer enhancement of dielectric fluid in a double-pipe annulus

M. Amirhakimi, ..., G. Ahmadibeni

Journal of Electrostatics • Volume 136 • 2025 • Article 104126

View PDF

Research article Open access

Deep reinforcement learning-based airfoil design and optimization: An aerodynamic analysis

P. Scavella, ..., C.S. Greco

Aerospace Science and Technology • Volume 167 • 2025 • Article 110638

View PDF

Research article Full text access

Designing a digital engineering curriculum for manufacturing education: industry-aligned course development

Ana Wooley

Manufacturing Letters • Volume 46 • 2025 • Pages 34-36

View PDF

Research article Full text access

Time series prediction for cryptocurrency markets with transformer and parallel convolutional neural networks

Mohammad Ali Izadi, Ehsan Hajizadeh

Applied Soft Computing • Volume 177 • 2025 • Article 113229

View PDF